LakefillQ: Purely Functional Data Structures, Substructural Typing, and Graded Physical Compilation for Low-Width Fault-Tolerant Quantum Circuits

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Quantum algorithms cannot overwrite live registers in place without entangling computational state with garbage ancillas and destroying interference. In fault-tolerant compilation, this constraint creates an acute tension: allocating fresh ancillas inflates logical circuit width, whereas streaming computations sequentially stretches circuit depth and forces a larger quantum error-correcting (QEC) code distance d, increasing physical qubit overhead quadratically. Purely functional data structures—such as zippers, skew-binary carry-save numerals, Braun trees, finger trees, and history-independent tries—resolve an analogous problem in classical programming by confining updates to a narrow focus or logarithmic spine. Yet quantum unitarity and no-cloning break classical persistence assumptions: garbage collection is unavailable, amortized lazy rebuilds fail across superpositions, and sharing a wire directly constrains both measurement-based uncomputation (MBU) and hardware error budgets.

We present LakefillQ, a functional quantum programming language and hardware-graded compiler that brings purely functional data structures to fault-tolerant quantum circuits without hardware QRAM assumptions. Our central insight is that wire sharing and physical resource consumption must be governed by a single compositional typing judgment, $\Gamma_{\text{in}} \vdash t : Q_e(A) \dashv \Gamma_{\text{out}}$, pairing a five-point ordered duoidal wire-capability algebra ($0 < \mathsf{BorrowZ} < \mathsf{TaintedX} < \mathsf{Lin1} < \mathsf{Uniq1}$) with a hardware-graded cost vector $e = (W_{\text{peak}}, W_{\text{clean}}, \Phi, \alpha, \varepsilon, E_{\text{ebit}})$ and nonlinear repetition/partition action $r \triangleright e$. Without user proof annotations, LakefillQ discharges domain-restricted erasure, quotient canonicality, and bilinear X-basis phase-taint obligations via XOR-AND graph equivalence sweeping, SMT solving, and $O(1)$-query k-inductive loop-invariant inference, lowers programs through non-Clifford ZX phase-polynomial e-graph saturation and chordal register coloring, and emits dynamic OpenQASM 3.0 and Stim circuits across single-machine, parallel multi-QPU, and serial multishot regimes.

Across 60 compiled benchmarks, LakefillQ establishes new Pareto frontiers on open problems: 256-bit elliptic-curve discrete logarithms in 1,088 logical qubits and 14.34×10^6 Toffolis; RSA-2048 factoring in 1,364 logical qubits without incurring the $\Delta d = +2$ code-distance penalty of sequential carry rippling; a 10.46× width reduction (to 9,984 qubits) for coherent Decoded Quantum Interferometry (DQI), extended to non-binary Hermitian, Suzuki, and lifted bivariate algebraic-geometry codes exceeding Reed–Solomon block-length bounds; and a 22.48× state-preparation Toffoli reduction for FeMoCo tensor-hypercontracted QSVT. All metatheory, data-structure invariants, and lowering passes are mechanized in 2,335 Lean 4.35.0 theorems across 84 modules with zero axioms or incomplete proofs.

1 INTRODUCTION

A central challenge in fault-tolerant quantum compilation is navigating the trade-off between logical circuit width, Toffoli gate volume, and physical error-correction overhead. By Bennett's reversibility principle [11], overwriting a live quantum register without uncomputing its prior value entangles computational basis states with residual ancillas and collapses interference. Naive out-of-place compilation avoids overwriting by allocating fresh ancillas at every step, causing width to grow linearly with circuit depth until uncomputation completes. Conversely, squeezing an algorithm into minimal width by streaming data sequentially through a small active register stretches execution depth—and in surface-code and quantum low-density parity-check (qLDPC) architectures [22, 72], a 32× increase in circuit runtime can force the QEC code distance d upward by $\Delta d = +2$, inflating physical qubits per logical patch and erasing much of the logical width savings.

Purely functional data structures [21, 52, 56, 82, 84] address a closely related non-overwriting discipline in classical software. Because persistent functional updates cannot mutate aliased nodes in place, structures such as Huet's zipper [56], Myers' skew-binary numerals [82], Braun trees [21,

83], finger trees [52], and Morrison's crit-bit tries [81] organize state so that updates touch only a bounded $O(1)$ focus or an $O(\log N)$ spine while leaving surrounding context untouched.

Translating classical persistent data structures into fault-tolerant quantum circuits, however, requires far more than syntactic rewriting. Three physical barriers separate unitary hardware from a classical functional runtime:

(1) **Explicit Uncomputation and Worst-Case Synchrony.** Quantum hardware has no garbage collector [12, 17, 88]. Every temporary wire must be returned to $|0\rangle$ either unitarily or via Gidney's X-basis measurement-based uncomputation (MBU) [38]. Moreover, amortized structures that occasionally trigger $O(N)$ lazy rebuilds [53] fail in superposition because a quantum circuit must execute the worst-case branch depth across all computational paths.
(2) **History Independence and QRAM-Free Access.** In superposition, two execution paths that construct the same logical set in different insertion orders fail to interfere if internal tree balance bits or freelist pointers retain history [14, 114]. Furthermore, without speculative hardware QRAM, lookups and shifts must compile to data-independent topologies such as Select-SWAP (QROAM) [9, 75], Batcher bitonic networks, or Variable-Time Amplitude Amplification (VTAA) [5].
(3) **Inseparability of Wire Sharing and Physical Cost.** Okasaki's classical persistence framework [84] treats structural pointer aliasing and amortized operational cost as independent concerns. Under no-cloning and hardware noise, they become tightly coupled: borrowing a wire read-only in the Z basis permits zero-Toffoli MBU only when the computed Boolean function has algebraic degree at most 2 over $\mathbb{F}_2$; mid-circuit X-basis measurements inject dynamic phase kickbacks that must be canceled before interference; and bursty MBU measurements can saturate an external QEC syndrome decoder if measurement arrival exceeds decoder throughput [106].

Our Insight and Approach. Our key insight is that purely functional structural locality and fault-tolerant quantum compilation are unified when wire capabilities and physical hardware grades live in one compositional type judgment:

$$\Gamma_{\text{in}} \vdash t : Q_e(A) \dashv \Gamma_{\text{out}}.$$

Rather than treating uncomputation safety [17], entanglement purity [115], canonical data representations [114], block-encoding error [33], and hardware layout as disconnected passes, λ_{LFQ} tracks a five-point ordered partial duoidal capability on every wire ($0 < \mathsf{BorrowZ} < \mathsf{TaintedX} < \mathsf{Lin1} < \mathsf{Uniq1}$) alongside a six-component resource grade $e = (W_{\text{peak}}, W_{\text{clean}}, \Phi, \alpha, \varepsilon, E_{\text{ebit}})$ acted on by a modal repetition/partition operator $r \triangleright e$. This design lets programmers write high-level functional abstractions—zippers, balanced tree folds, carry-save limbs, and asymmetric MBU blocks—in `#lang lakefillq` with zero manual proof annotations, while the compiler automatically proves phase-taint cancellation, recycles dead wires on chordal Static Single Assignment (SSA) graphs, and extracts target-optimal gate schedules.

Contributions. This paper makes four main contributions:

(1) **A Unified Substructural and Hardware-Graded Quantum Calculus (§2–§3).** We formalize λ_{LFQ} and its six-primitive kernel $\lambda_{\text{LFQ}}^{\text{kern}}$, showing how domain-restricted erasure (U1), quotient canonicality (U2), orthogonal branching (U3), bilinear X-basis phase cancellation (U4), hardware error/decoder/crosstalk grades (Φ_1–Φ_3), Chebyshev block-encoding fusion (QBlock), and three-span error diagnostics all arise as projections of $\Gamma_{\text{in}} \vdash t : Q_e(A) \dashv \Gamma_{\text{out}}$,

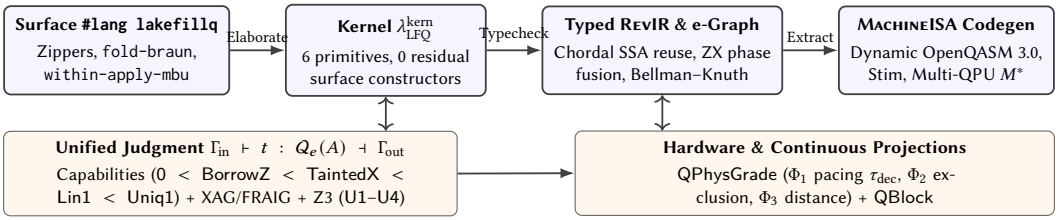


Fig. 1. Architecture of the LakefillQ compiler (LakefillQ → RevIR → MachineISA). All logical uncomputation laws (U1–U4), hardware grades (Φ_1–Φ_3), block-encoding bounds (QBlock), and three-span diagnostics project from one typing judgment.

scaled to multi-round ciphers via automatic XOR-AND graph (XAG/FRAIG) cut-sweeping and $O(1)$-query k-inductive loop-invariant inference.

(2) **Purely Functional Quantum Data Structures (§4).** Grounded in two sharp uncomputation laws—an exact degree-2 characterization of Clifford MBU (Proposition 4.1) and an $h+1$ register bound for two-pass tree pebbling (Proposition 4.2)—we design nine core and four bootstrapped functional quantum data structures in `#lang lakefillq` that confine non-Clifford work to narrow foci, decouple carry chains, and guarantee history-independent state representations without QRAM.

(3) **ZX Phase-Polynomial e-Graph Lowering and Multi-QPU Partitioning (§5).** We build a three-dialect compiler (LakefillQ → RevIR → MachineISA) combining chordal SSA wire reuse, congruence-closed e-graph saturation with non-Clifford ZX phase-polynomial spider fusion and shared-control twin-CCZ rules [65], Bellman–Knuth hardware extraction across four QEC/NISQ profiles, 64-bit bitboard Meet-in-the-Middle (MITM) leaf synthesis, and convex $M^* \leq 256$ multi-QPU partitioning with Law-U4 GHZ cat-disentangling.

(4) **Empirical Validation, Open-Problem Advances, and Mechanized Proofs (§6).** Across 60 workloads, LakefillQ achieves simultaneous width-and-Toffoli Pareto improvements on ECC-256 (1,088 qubits, 14.34×10^6 Toffolis), code-distance-preserving RSA-2048/4096 factoring ($d = 27$ vs. $d = 29$), 10.46× narrower OPI DQI plus non-binary Hermitian, Suzuki, and lifted bivariate Early-MBU DQI beating Reed–Solomon length bounds, and 22.48× cheaper FeMoCo QSVT uncomputation. We further confirm across nine third-party quantum libraries (14 concrete violations) that λ_{LFQ}'s side conditions rule out real-world circuit hazards rather than artificial pathologies, pass six-engine differential fuzzing and 12/12 mutation kills, and mechanize all metatheory in 2,335 zero-axiom Lean 4.35.0 theorems across 84 modules.

2 OVERVIEW: PROGRAMMING AND COMPILING IN LAKEFILLQ

Before formalizing λ_{LFQ}, we walk through two concrete programs in `#lang lakefillq` that illustrate why naive reversible circuits either waste Toffoli gates or silently corrupt quantum state, and how LakefillQ resolves both hazards statically.

2.1 Asymmetric Uncomputation and Focus Isolation: Modulo-13 Addition

Consider computing $|a, b, \text{out}\rangle \mapsto |a, b, \text{out} \oplus ((a+b) \bmod 13)\rangle$ on 4-bit integers using nine scratch ancillas for ripple carries and modulo-13 Boolean reduction. Standard Bennett compute–copy–uncompute (`within-apply` in Silq [17] or Qrisp [102]) computes nine forward Toffolis (`TempANDs`), XORs the reduced bits into `out`, and mirrors nine reverse Toffolis to return the scratch wires to $|0\rangle$—spending 18 Toffolis total.

Listing 1 expresses this circuit in `#lang lakefillq` using 9 Toffolis instead of 18. Two language abstractions work in concert:

- **Scoped Focus (`focus-zipper`).** Wrapping the output accumulation in `(focus-zipper [out-z out (append a b out)] ...)` grants linear mutable capability (Lin1) exclusively to `out` while demoting the surrounding context `a`, `b`, and the live scratch wires `and9` to read-only Z-basis borrows (BorrowZ). Any accidental CNOT or Toffoli targeting `a` or `and9` inside the action block is rejected at compile time with a three-span diagnostic.
- **Asymmetric Measurement-Based Uncomputation (`within-apply-mbu`).** Instead of mirroring nine reverse Toffolis, `within-apply-mbu` lowers the reverse pass by measuring each scratch wire u_k in the X basis (H followed by Z-measurement $m_k \in \{0, 1\}$ and reset to $|0\rangle$). Measuring $|u_k\rangle = |f_k(x) \wedge g_k(x)\rangle$ in the X basis injects a state-dependent phase $(-1)^{m_k(f_k(x) \wedge g_k(x))}$, promoting the control wires to capability TaintedX. The compiler symbolically extracts the exact linear fan-in parity cones f_k, g_k over $\mathbb{F}_2$ and emits classical feed-forward `if (m[k] == 1) cz` fixups that cancel every quadratic phase term, restoring BorrowZ with zero reverse Toffolis.

Listing 1. Modulo-13 adder in `#lang lakefillq` (`examples/mod13_adder_mbu.lfq`): `focus-zipper` isolates mutation to `out`, while `within-apply-mbu` halves TempAND count from 18 to 9 with zero annotations.

```
#lang lakefillq
(target-profile heron-r2)

(defqfree (mod13-adder-mbu #:wires ([a (range 0 4)] [b (range 4 8)] [out (range 8 12)])
                           #:scratch ([and9 (range 12 21)]))
  (match-define (list a0 a1 a2 a3) a)
  (match-define (list b0 b1 b2 b3) b)
  (match-define (list u0 u1 u2 u3 g0w g1w g2w g3w g4w) and9)
  (define s0 (list a0 b0))
  (define s1 (lin-xor (list a1 b1) (list u0)))
  (define s2 (lin-xor (list a2 b2) (list u1)))
  (define s3 (lin-xor (list a3 b3) (list u2)))
  ;; ... linear XOR definitions r0-form .. r3-form elided for brevity ...
  (within-apply-mbu
   (seq ;; Compute block: 4 ripple-carry + 5 mod-13 reduction TempANDs (9 forward, 0 reverse)
    (temp-and a0 b0 u0)
    (temp-and-lin (lin-xor (list a1) (list u0)) (lin-xor (list b1) (list u0)) u1) (cx u0 u1)
    (temp-and-lin (lin-xor (list a2) (list u1)) (lin-xor (list b2) (list u1)) u2) (cx u1 u2)
    (temp-and-lin (lin-xor (list a3) (list u2)) (lin-xor (list b3) (list u2)) u3) (cx u2 u3)
    (temp-and-lin s3 (lin-xor s2 s3) g0w)
    (temp-and-lin (lin-xor s3 (list u3) (list g0w)) s0 g1w #t #t)
    (temp-and-lin (lin-xor s1 (list g1w)) (lin-xor s0 s1) g2w #t #f)
    (temp-and-lin (lin-xor s1 s3 (list u3 g0w g1w g2w)) (lin-xor s0 s1 s2 (list g1w g2w)) g3w #t #t)
    (temp-and-lin (lin-xor s0 s1 s3 (list g0w g2w g3w)) (lin-xor s1 s2 s3 (list u3 g0w g1w g2w)) g4w #t #f))
   (focus-zipper [out-z out (append a b out)] ;; Action block: only out-z is mutable (Lin1)
     (for/seq ([w (in-list r0-form)]) (cx w (wref out-z 0)))
     (x (wref out-z 1)) (for/seq ([w (in-list r1-form)]) (cx w (wref out-z 1)))
     (x (wref out-z 2)) (for/seq ([w (in-list r2-form)]) (cx w (wref out-z 2)))
     (x (wref out-z 3)) (for/seq ([w (in-list r3-form)]) (cx w (wref out-z 3))))
   and9))
```

What goes wrong without our type system? First, if a programmer applies H to an input wire while a TaintedX phase kickback $(-1)^{m_k(a_0 \wedge b_0)}$ remains uncanceled, Hadamard interference mixes the wrong signs and drops state fidelity to ≤ 0.50; Rule T-PhaseClean rejects the program and exhibits a minimal basis counterexample $|a_0 = 1, b_0 = 1\rangle$ (Listing 2). Second, on hardware where readout error exceeds twice the two-qubit gate error ($\epsilon_{\text{meas}} > 2\epsilon_{2Q}$) or where an external QEC syndrome decoder with latency τ_{dec} cannot keep pace with bursty MBU resets, MBU actually degrades fidelity; as Figure 1 shows, LakefillQ's QPhysGrade e-graph extractor automatically selects between MBU, measurement-free Margolus RCCX, or native Rydberg CCZ per target profile.

Listing 2. Three-span resugared diagnostic emitted by `lakefillqc --diagnostics-human` when Hadamard interference is attempted on a wire carrying uncanceled X-basis phase taint across a chordally recycled ancilla.

```
error[PhaseTaint::LFQ-E-U4-TAINT-INTERFERE]: gate `h` applied to `a[0]` with TaintedX
  origin_span --> mod13_buggy.lfq:18:4 `(temp-and a0 b0 u0)` [recycled q[12] <- prev_s[0]]
  mutating_span --> mod13_buggy.lfq:29:4 `(mid-meas-x u0 m0)` [phase (-1)^(m0 * (a0 & b0))]
  failure_span --> mod13_buggy.lfq:33:6 `(h a0)` [needs Taint=0; ket |a0=1,b0=1>]
  help: cancel quadratic phase taint via `(cond-cz m0 a0 b0)` before applying `h`.
```

2.2 Domain-Restricted Erasure and Balanced Folds: Non-Binary Hermitian AG DQI

Whereas modulo-13 addition is total on all 4-bit inputs and uncomputes scratch wires at block exit, our second example—Decoded Quantum Interferometry (DQI) [60]—requires domain-restricted erasure (U1), early error-register reclamation (`mid-meas-x`), and balanced tree folds (`fold-braun`). Standard DQI reduces Max-LINSAT over a finite field $\mathbb{F}_q$ to coherently decoding a dual linear code $[n, k, d]_{\mathbb{F}_q}$ in superposition: after encoding an error state $|e\rangle$ into its syndrome $|He\rangle$, the circuit must uncompute $|e\rangle \mapsto |0\rangle$ before applying a quantum Fourier/Hadamard transform to $|He\rangle$. Over $\mathbb{F}_4$, classical Reed–Solomon codes suffer from the Maximum Distance Separable (MDS) blocklength ceiling $n_{\mathrm{RS}} \leq q + 1 = 5$, forcing larger field alphabets to reach longer blocklengths. Algebraic-geometry (AG) codes on curves of genus $g \geq 1$ [35, 105] beat this ceiling: the genus-1 Hermitian curve $H_2 : x^3 = y^2 + y$ over $\mathbb{F}_4$ has $n = q^{3/2} = 8 > 5$ rational affine points, yielding an $[8, 5, 3]_{\mathbb{F}_4}$ code on 16 error qubits and 6 syndrome qubits (22 qubits total).

Listing 3. Early-MBU DQI on the $[8, 5, 3]_{\mathbb{F}_4}$ Hermitian curve $H_2 : x^3 = y^2 + y$ (`examples/early_mbu_dqi_hermitian_ag_f4.lfq`), shrinking live width from 22 to 6 qubits via `mid-meas-x` and `fold-braun`.

```
#lang lakefillq
(target-profile bb-qldpc)
(defconst hermitian-h2-points '((0 0) (0 1) (1 2) (1 3) (2 2) (2 3) (3 2) (3 3)))

(defq (early-mbu-dqi-hermitian-ag-f4
       #:wires ([e16 (range 0 16)] [syn6 (range 16 22)])
       #:scratch ([v-t0 (range 22 30)] [v-t1 (range 30 38)] [v-loc (range 38 46)])
       #:domain (domain-guard hermitian-f4-wt-le1-guard))
  ;; 1. Encode GF(4)^3 Hermitian parity checks syn6 ^= H_{H2} * e16
  (for/seq ([j (in-range 8)])
    (match-define (list xj yj) (list-ref hermitian-h2-points j))
    (seq (gf4-mac 1  (wref e16 (* 2 j)) (wref e16 (add1 (* 2 j))) (wref syn6 0) (wref syn6 1))
         (gf4-mac xj (wref e16 (* 2 j)) (wref e16 (add1 (* 2 j))) (wref syn6 2) (wref syn6 3))
         (gf4-mac yj (wref e16 (* 2 j)) (wref e16 (add1 (* 2 j))) (wref syn6 4) (wref syn6 5))))
  ;; 2. Early-MBU: measure & reset all 16 error wires immediately (22 -> 6 live qubits!)
  (for/seq ([j (in-range 8)])
    (match-define (list m0 m1) (list-ref hermitian-h2-meas-syms j))
    (seq (mid-meas-x (wref e16 (* 2 j)) m0) (mid-meas-x (wref e16 (add1 (* 2 j))) m1)))
  ;; 3. Braun-tree curve-locator phase cancellation across all 24 non-zero coset leaders
  (fold-braun
   (lambda (pt j)
     (match-define (list xj yj) pt)
     (match-define (list m0 m1) (list-ref hermitian-h2-meas-syms j))
     (within-apply-mbu
      (seq (gf4-mac xj (wref syn6 0) (wref syn6 1) (wref syn6 2) (wref syn6 3))
           (gf4-mac yj (wref syn6 0) (wref syn6 1) (wref syn6 4) (wref syn6 5))
           (x (wref syn6 2)) (x (wref syn6 3)) (x (wref syn6 4)) (x (wref syn6 5))
           (temp-and (wref syn6 2) (wref syn6 3) (wref v-t0 j))
           (temp-and (wref syn6 4) (wref syn6 5) (wref v-t1 j))
           (temp-and (wref v-t0 j) (wref v-t1 j) (wref v-loc j)))
      (seq (cond-cz m0 (wref syn6 0) (wref v-loc j))
           (cond-cz m1 (wref syn6 1) (wref v-loc j)))
      (list (wref v-loc j) (wref v-t1 j) (wref v-t0 j))))
   hermitian-h2-points)
  ;; 4. Interference on the 6-qubit syndrome register (verified clean of Law-U4 phase taint)
  (focus-zipper [sz syn6 (append e16 syn6)]
    (for/seq ([r (in-range 6)]) (h (wref sz r)))))
```

In unitary DQI, all 16 error qubits must stay alive alongside the syndrome decoder. In Listing 3, LakefillQ's Early-MBU idiom measures and frees all 16 error wires *immediately* after syndrome encoding via `mid-meas-x`, collapsing live logical width from 22 down to 6 qubits (9 active qubits including 3 chordally recycled scratch wires across the 8 iterations of `fold-braun`) and cutting T-gates by 3.5× (336 → 96, with zero reverse Toffolis). Notice that syndrome decoding is only injective on the domain $\mathcal{D}$ of error vectors of $\mathbb{F}_4$-symbol weight ≤ 1 (25 coset leaders). Without the `domain-guard`, decoding would fail unconditionally on weight-2 states; with `domain-guard`, LakefillQ automatically escalates from sparse ANF to XAG/FRAIG + Z3 equivalence checking and proves in 0.18 s that every dynamic phase kickback $(-1)^{\langle m,e\rangle}$ cancels identically across all 25 coset leaders before the final Hadamards execute. Section 6.1 (Table 6) shows that this same Early-MBU curve-locator pattern scales to lifted bivariate codes ($\mathbb{F}_4^2$), higher-genus Hermitian curves (H_3, H_4, H_8), and Deligne–Lusztig Suzuki curves ($S(8)$).

3 CORE CALCULUS AND THE QUANTUM OKASAKI TYPE SYSTEM

Existing quantum type systems each enforce one facet of correctness: Silq [17] checks uncomputation syntactically via `qfree`/`mfree`, Twist [115] tracks entanglement purity, Tower [114] lowers classical pointer structures unitarily, Proto-Quipper-R/QuRA [27] bounds peak width via affine grades, and Cobble [33] tracks block-encoding subnormalization. On fault-tolerant hardware, however, these properties cannot be checked in isolation: an X-basis MBU reset leaves wires logically unentangled in the Z basis yet tainted by dynamic phase kickbacks (U4), and whether that reset beats unitary uncomputation depends on readout error and external QEC syndrome-decoder latency (Φ_1). LakefillQ captures discrete wires, continuous block-encodings, and hardware resources in one typing judgment.

3.1 Kernel Calculus, Wire Capabilities, and Cost Grades

Surface-to-Kernel Elaboration. A `#lang lakefillq` module comprises declarations (`defq`, `defqfree`, `defconst`, `target-profile`) built from 20 functional surface forms, including scoped zippers (`focus-zipper`), asymmetric uncomputation (`within-apply-mbu`), balanced tree catamorphisms (`fold-braun`), bounded inductive loops (`for/fold-loop`), linear XOR fan-ins (`lin-xor`, `temp-and-lin`), modular subcircuits (`call-q`), Chebyshev QSVT walks (`qsvt-cheb`), and cluster partitions (`par-cluster`). Elaboration compiles all 20 forms into a six-primitive substructural kernel $\lambda_{\mathrm{LFQ}}^{\mathrm{kern}}$:

$$k \;::=\; \mathsf{skip} \;\mid\; \mathsf{gate}(g, \vec{w}) \;\mid\; k_1 \,;; k_2 \;\mid\; k_1 \otimes k_2 \;\mid\; \mathsf{mid\text{-}meas\text{-}x}(w, m) \;\mid\; \mathsf{cond\text{-}pauli}(\phi_{\mathbb{F}_2}(\vec{m}), P, \vec{w}),$$

where $\mathcal{G}_{\mathrm{phys}} = \{\mathrm{X}, \mathrm{CNOT}, \mathrm{CCX}, \mathrm{CSWAP}, \mathrm{H}, \mathrm{S}, \mathrm{S}^\dagger, \mathrm{MEAS_X_PAULI_FRAME}\}$ requires zero hardware QRAM [12, 38]. Because purely functional bindings are natively in Static Single Assignment (SSA) form, their live-range interference graphs are chordal ($\chi(G) = \omega(G) = \max_t |\mathrm{Live}(t)|$) [46, 93], making linear-scan clean-ancilla reuse exact.

Theorem 3.1 (Sound Elaboration and Chordal Clean-Ancilla Reuse)**.** *Elaboration translates every well-typed* `#lang lakefillq` *program into* $\lambda_{\mathrm{LFQ}}^{\mathrm{kern}}$ *with zero residual surface constructors. Lowering* $\mathsf{within\text{-}apply\text{-}mbu}(C, A, \vec{u})$ *emits* m *forward* `TempAND`*s and zero reverse Toffolis while preserving exact operational equivalence with Bennett uncomputation, and chordal SSA wire coloring reuses dead* $|0\rangle$ *ancillas without aliasing live state.*

Ordered Partial Duoidal Wire Capabilities. Every typing context Γ maps each wire w to a capability from a five-point ordered algebra (annotated with chiplet place $@\,p$ in distributed mode):

$$0 \;<\; \mathsf{BorrowZ} \;<\; \mathsf{TaintedX} \;<\; \mathsf{Lin1} \;<\; \mathsf{Uniq1}.$$

$$\frac{\Gamma_F : \mathsf{Lin1}^{|F|} \quad \Gamma_C : \mathsf{BorrowZ}^{|C|} \quad \Gamma_F, \Gamma_C \vdash t : \mathcal{Q}_e(A) \dashv \Gamma'_F, \Gamma_C}{\Gamma_F, \Gamma_C \vdash (\texttt{focus-zipper}\ \ [z\ F\ (F \cup C)]\ t) : \mathcal{Q}_e(A) \dashv \Gamma'_F, \Gamma_C}\ \text{[T-FocusZipper]}$$

$$\frac{u : \mathsf{Uniq1} \in \Gamma \quad \mathrm{ANF}(u) = f(\vec{x}) \wedge g(\vec{x}) \quad \vec{x} : \mathsf{BorrowZ} \in \Gamma}{\Gamma \vdash (\texttt{mid-meas-x}\ u\ m) : \mathcal{Q}_{(1,1,\Phi_{\mathrm{meas}},1,0,0)}(\mathsf{Bit}_m) \dashv \Gamma[u \mapsto 0,\ \vec{x} \mapsto \mathsf{TaintedX}^{\,m\cdot(f\wedge g)}]}\ \text{[T-MidMeasX]}$$

$$\frac{\vec{x} : \mathsf{TaintedX}^{\,\psi+m\cdot(f\wedge g)} \in \Gamma \quad \mathrm{ANF}(w_1) = f(\vec{x}) \quad \mathrm{ANF}(w_2) = g(\vec{x})}{\Gamma \vdash (\texttt{cond-cz}\ m\ w_1\ w_2) : \mathcal{Q}_{(0,0,\Phi_{\mathrm{cz}},1,0,0)}(\mathsf{Unit}) \dashv \Gamma[\vec{x} \mapsto \mathsf{TaintedX}^{\,\psi}]}\ \text{[T-CondCZ]}$$

$$\frac{w : \kappa \in \Gamma \quad \kappa \in \{\mathsf{Lin1}, \mathsf{Uniq1}\} \quad \mathrm{Taint}(w) \equiv_{\mathcal{D}} 0}{\Gamma \vdash \mathsf{gate}(H, w) : \mathcal{Q}_{(1,0,\Phi_H,1,0,0)}(\mathsf{Unit}) \dashv \Gamma}\ \text{[T-PhaseClean]} \qquad \frac{\forall x \in \mathcal{D}.\ g(x, f(x)) = x \quad \vec{u}_{\mathrm{out}} \equiv_{\mathcal{D}} 0}{\Gamma \vdash \mathsf{erase}_{\mathcal{D}}(f, g, \vec{u}) : \mathcal{Q}_e(\mathsf{Unit}) \dashv \Gamma[\vec{u} \mapsto 0]}\ \text{[T-DomImg]}$$

$$\frac{\Gamma, \vec{u} : 0 \vdash C : \mathcal{Q}_{e_C}(\mathsf{Unit}) \dashv \Gamma_1, \vec{u} : \mathsf{Uniq1} \quad \Gamma_1, \vec{u} : \mathsf{BorrowZ} \vdash A : \mathcal{Q}_{e_A}(B) \dashv \Gamma_2, \vec{u} : \mathsf{BorrowZ} \quad \mathrm{Deg}_{\mathbb{F}_2}(u_i) \le 2}{\Gamma \vdash (\texttt{within-apply-mbu}\ C\ A\ \vec{u}) : \mathcal{Q}_{e_C ;; e_A ;; e_{\mathrm{mbu}}}(B) \dashv \Gamma_2}\ \text{[T-WithinMBU]}$$

$$\frac{\Gamma \vdash U : \mathsf{QBlock}(A, \alpha, m_{\mathrm{anc}}, \varepsilon_0, \Phi_X) \dashv \Gamma \quad (d \text{ even} \implies U^2 = I) \quad \Phi_X = 0}{\Gamma \vdash (\texttt{qsvt-cheb}\ d\ U) : \mathcal{Q}_{\mathsf{cheb}(d) \triangleright e_U}\big(\mathsf{QBlock}(T_d(A/\alpha), 1, m_{\mathrm{anc}} + 1, 4d\varepsilon_0, 0)\big) \dashv \Gamma}\ \text{[T-QSVTFused]}$$

$$\frac{M^* = \arg\min_{M \in \{1..M_{\max}\}} \mathrm{Vol}_{\mathrm{phys}}(\mathsf{par}(M) \triangleright e) \quad \forall p.\ W_{\mathrm{loc}}(p) \le Q_{\mathrm{cap}}(p)}{\Gamma \vdash (\texttt{par-cluster}\ M^*\ t) : \mathcal{Q}_{\mathsf{par}(M^*) \triangleright e}(A) \dashv \Gamma'}\ \text{[T-ParGuard]}$$

Fig. 2. Selected syntax-directed typing rules of λ_{LFQ} projecting from $\Gamma_{\mathrm{in}} \vdash t : \mathcal{Q}_e(A) \dashv \Gamma_{\mathrm{out}}$.

Operationally, 0 denotes an absent or $|0\rangle$-reclaimed wire; BorrowZ permits read-only Z-basis control sharing ($\Delta_Z : \mathsf{Lin1} \to \mathsf{BorrowZ} \otimes \mathsf{BorrowZ}$); TaintedX marks a wire carrying an uncanceled classical-measurement phase $(-1)^{\phi(m,x)}$ that commutes with diagonal Z-controls yet forbids basis-mixing gates (H, X, CSWAP) or scope exit until canceled; Lin1 grants linear unitary mutation; and Uniq1 grants exclusive ownership required for destructive reset or initialization [77]. Sequential composition ($;;$) and parallel tensor ($\otimes$) form an ordered partial duoidal algebra satisfying lax interchange $(a \otimes b) ;; (c \otimes d) \sqsubseteq (a ;; c) \otimes (b ;; d)$ across all $5^4 = 625$ capability quadruples: a wire may be borrowed twice in parallel and then linearly mutated sequentially, whereas parallel linear/borrow aliasing ($\mathsf{Lin1} \otimes \mathsf{BorrowZ}$) is undefined. Disjoint array types (`qvec n`) and modular (`call-q ...`) calls contract shared BorrowZ controls via Δ_Z while rejecting aliased mutable ports.

Hardware-Graded Cost Vector and Modal Action. Each term $\Gamma_{\mathrm{in}} \vdash t : \mathcal{Q}_e(A) \dashv \Gamma_{\mathrm{out}}$ carries a six-component grade

$$e = \big(W_{\mathrm{peak}},\ W_{\mathrm{clean}},\ \Phi,\ \alpha,\ \varepsilon,\ E_{\mathrm{ebit}}\big),$$

tracking peak live width W_{peak}, reclaimed $|0\rangle$ ancillas W_{clean}, hardware grade $\Phi \in \mathsf{QPhysGrade}$, block-encoding subnormalization $\alpha \ge 1$, spectral error $\varepsilon \ge 0$, and inter-QPU Bell pairs E_{ebit}. Sequential composition reuses clean ancillas directly:

$$W_{\mathrm{peak}}(e_1 ;; e_2) = \max\big(W_{\mathrm{peak}}(e_1),\ (W_{\mathrm{peak}}(e_1) - W_{\mathrm{clean}}(e_1)) + W_{\mathrm{peak}}(e_2)\big), \quad \varepsilon(e_1 ;; e_2) = \varepsilon_1 + \varepsilon_2.$$

Bounded loop iteration $R \triangleright e_{\mathrm{step}}$ and two genuinely quantum nonlinearities enter through a single modal operator $r \triangleright e$ [37, 86]: degree-d Chebyshev QSVT iteration $\mathsf{cheb}(d) \triangleright e$ fuses reflection ancillas ($W_{\mathrm{clean}} \mapsto m_{\mathrm{anc}}+1$) while accounting for quadratic coherent phase-angle drift ($4d\varepsilon_0 + d^2\varepsilon_{\mathrm{rot}}^2$, reducing to $4d\varepsilon_0$ under exact dyadic catalyst angles where $\varepsilon_{\mathrm{rot}} = 0$), and M-way cluster partitioning $\mathsf{par}(M) \triangleright e$ scales per-QPU width ($\lceil W_{\mathrm{loc}}/M \rceil + 2E_{\mathrm{cut}}$) while charging Zalka's $\sqrt{M}$ Grover work factor [116] on unstructured search (`'auto` derives cluster parameters directly from the typed AST without user metadata). Figure 2 displays the core typing rules; seven static projections (T-DomImg, T-BorrowZ, T-PhaseClean, T-CounitU4, T-SubNorm, T-QubitCap, T-ParGuard) enforce every language invariant, and §6.5 verifies empirically that each side condition rules out concrete hazards observed in third-party quantum circuit libraries.

3.2 Logical Projection (QSub): Uncomputation Laws and Zero-Flag XAG/FRAIG Scaling

The logical projection QSub prevents four distinct bugs that cause silent entanglement or destructive interference in quantum circuits:

- **Law U1 (Domain-Restricted Inverse Erasure, T-DomImg).** Algebraic decoders (BCH, Reed–Solomon, Hermitian AG) and Montgomery/Kaliski inverters only invert f on a subset $\mathcal{D} \subset \{0,1\}^n$ (such as bounded-weight error cosets or odd residues). Requiring global invertibility rejects valid algorithms, whereas omitting the guard leaves dirty ancillas outside $\mathcal{D}$. Law U1 checks $g(x, f(x)) = x$ restricted to the explicit `domain-guard` predicate $\mathcal{D}$.
- **Law U2 (Quotient Canonicality, T-PhaseClean).** Carry-save limbs, RNS representations, and search trees admit multiple bit encodings for one mathematical value. Before Hadamard/QFT interference or state comparison, Law U2 verifies that every value in $\mathcal{D}$ maps to a unique canonical representative.
- **Law U3 (Exhaustive Orthogonal Branching).** Reversible pattern matches and VTAA steps must span mutually orthogonal, exhaustive subspaces on $\mathcal{D}$ lest amplitudes leak or double-count.
- **Law U4 (Bilinear $\mathbb{F}_2$ Phase-Taint Cancellation, T-CounitU4).** Mid-circuit X-basis resets accumulate symbolic phase polynomials $\Phi_X(m, x) \in \mathbb{F}_2[m, x]$; every measurement variable m_k must have identically zero net coefficient modulo $\mathcal{D}$ before scope exit or basis mixing.

Automatic Escalation from Sparse ANF to XAG/FRAIG + Z3. Programmers write zero proof annotations. For shallow arithmetic and tree gadgets, LakefillQ evaluates exact wire-level Algebraic Normal Form (ANF) polynomials over $\mathbb{F}_2$ (36 ms in Racket across our verification corpus, cross-checked in 489 ms via Z3). On multi-round cryptographic permutations where explicit ANF degree grows exponentially (2^{128} monomials), the checker automatically detects when intermediate fan-in cones exceed 64 monomials and escalates—with zero CLI flags—to a hash-consed XOR-AND Graph (XAG) engine with structural FRAIG reductions and Z3 Tseitin cut-sweeping. Because each individual `TempAND` $u_k = f_k(x) \wedge g_k(x)$ has bilinear degree $(1, 1)$ in its immediate XAG inputs, full-cipher unrolled cut-sweeping discharges 10-round AES-128 (340 `TempANDs`) in 3.06 s, 14-round AES-256 (510 `TempANDs`) in 4.25 s, 18-round Keccak-$f[200]$ (450 `TempANDs`) in 9.32 s, and 64-bit Gidney ripple addition in 3.39 ms.

$O(1)$-Query k-Inductive Loop-Invariant Inference (`for/fold-loop`). When iterated rounds are written as bounded functional loops (`for/fold-loop`) rather than unrolled straight-line blocks, `infer_inductive_loop_qsub` verifies a single symbolic step $\Gamma_{\text{inv}} \vdash B_{\text{step}} : \mathcal{Q}_{e_{\text{step}}}(A) \dashv \Gamma_{\text{inv}}$ ($e_{\text{loop}} = R \triangleright e_{\text{step}}$) in $O(1)$ Z3 queries independent of round count R: checking per-round scratch erasure (U1), single-step permutation bijectivity (U2), and zero loop-carried TaintedX escape (U4) accelerates Z3 round-invariant solving over $O(R)$ unrolled round checks by 16.78× on AES-128 (5.75 ms vs. 96.52 ms; 3 vs. 30 Z3 calls), 15.70× on AES-256 (5.46 ms), 17.89× on Keccak-$f[200]$ (3.90 ms), 19.04× on 20-round ChaCha20 (3.79 ms), and 1,431.60× on $R = 1{,}000$ iterated permutations (2.99 ms vs. 4,278.42 ms; 2 vs. 2,000 Z3 calls), while deterministically rejecting 4/4 defective loop invariants.

Theorem 3.2 (Logical Preservation and Four-Way Independence of QSub)**.** *Any kernel term deriving $\Gamma_{\text{in}} \vdash t : \mathcal{Q}_e(A) \dashv \Gamma_{\text{out}}$ with empty residual TaintedX (including bounded loops certified via $O(1)$-query k-inductive QSub inference) preserves reduced state purity $\text{Tr}(\rho_{\text{out}}^2) = 1$ across all inputs in $\mathcal{D}$ and all 2^k mid-circuit M_X trajectories. Furthermore, Laws U1–U4 are strictly pairwise independent (Table 1).*

Table 1. Strict four-way independence of QSub Laws U1–U4 verified on statevector and density-matrix simulators: each witness satisfies three laws, violates exactly one, and collapses output purity or interference.

Violated Law	Minimal Witness	Laws OK	Physical Failure Mode	Purity / Overlap
U1 (T-DomImg)	Unguarded BCH$(15,7,5)$ wt-3	U2, U3, U4	Syndrome entangles data	$\mathrm{Tr}(\rho^2) = 0.5000$
U2 (Canonical)	Uncollapsed limb before QFT	U1, U3, U4	Split coset peaks ruin QFT	Overlap = 0.2500
U3 (Orthogonal)	Overlapping VTAA prefix guard	U1, U2, U4	Non-unitary amplitude leak	Norm = 0.7071
U4 (T-CounitU4)	Omitted cond-cz after M_X	U1, U2, U3	Random $(-1)^{m(a\wedge b)}$ phase	$\mathrm{Tr}(\rho^2) = 0.5000$

Table 2. Strict three-way independence of physical hardware grades Φ_1, Φ_2, and Φ_3 on noisy density-matrix and Stim fault-injection simulations (all witnesses satisfy QSub Laws U1–U4).

Grade	Hazard Witness	OK	Physical Failure	Measured Impact
Φ_1 (Pacing/T_1)	Unpaced τ_{dec}=4,500 ns MBU	Φ_2, Φ_3	Lindley FIFO backlog stall	Surv. 0.9995 $\to$ 0.5777
Φ_2 (Exclusion)	Parallel Rydberg gates $< R_b$	Φ_1, Φ_3	Dipolar blockade phase err.	Fid. 0.9921 $\to$ 0.6140
Φ_3 (Hook/Dyadic)	Unflagged hook syndrome tree	Φ_1, Φ_2	Wt-1 fault $\mapsto$ wt-2 data	d_{eff} : 5 $\to$ 3 (11.4×)

3.3 Physical Projection (QPhysGrade) and External Decoder Pacing

Even a circuit that is logically pure ($\mathrm{Tr}(\rho^2) = 1$ on an ideal simulator) fails on physical hardware if idle memory decoheres, bursty MBU measurements overwhelm the external QEC syndrome decoder, parallel gates trigger coupler crosstalk, or Clifford+T synthesis introduces hook errors. The hardware grade $\Phi = (\Phi_1, \Phi_2, \Phi_3) \in \mathsf{QPhysGrade}$ forms an ordered idempotent-parallel resource semiring tracking three independent physical hazards:

- **Grade Φ_1 (Lindblad Decoherence and Parameterized External Decoder Pacing τ_{dec}).** Tracks cumulative T_1/T_ϕ idling decay alongside a queueing bound against an *external* real-time QEC decoder characterized by its round latency parameter τ_{dec}. When a circuit emits MBU measurements at interval τ_{cycle}, Terhal's backlog theorem [106] dictates that if $\tau_{\mathrm{dec}} > \tau_{\mathrm{cycle}}$ (for instance, unpaced software MWPM at $\tau_{\mathrm{dec}} = 4{,}500$ ns vs. $\tau_{\mathrm{cycle}} = 1{,}000$ ns), the reaction-frame FIFO diverges linearly (reaching 146,205.0 ns over 80,000 rounds) and logical survival collapses as T-gates stall on delayed Pauli frames. Rather than hardcoding custom decoder hardware into the language, Φ_1 treats τ_{dec} as a target-profile parameter: whenever $\tau_{\mathrm{dec}} \le \tau_{\mathrm{cycle}}$ (satisfied by modern sub-microsecond FPGA/cryo-CMOS Union-Find decoders at $\tau_{\mathrm{dec}} \in [120, 420]$ ns; Appendix C), MBU lowering is enabled with zero FIFO backlog; when $\tau_{\mathrm{dec}} > \tau_{\mathrm{cycle}}$, LakefillQ automatically throttles measurement cadence or falls back to unitary uncomputation.
- **Grade Φ_2 (Coupler ZZ Crosstalk and Rydberg Blockade Exclusion).** Tracks simultaneous two-qubit conflict graphs on fixed-frequency transmon couplers and neutral-atom Rydberg blockade radii R_b, serializing or coloring conflicting parallel gates in $k_1 \otimes k_2$.
- **Grade Φ_3 (Hook-Safe Effective Code Distance and Dyadic Angle Exactness).** Tracks fault-polynomial weight so single mid-gadget ancilla faults cannot propagate into weight-2 data errors (halving effective distance d_{eff} from 5 to 3), while typing phase-gradient rotations over $\mathbb{Z}[1/2^k]$ to avoid online Ross–Selinger `gridsynth` T-count blowup [99].

Theorem 3.3 (Hardware Semiring Monotonicity and Three-Way Independence of QPhysGrade)**.** $(\mathsf{QPhysGrade}, ;, \otimes, \preceq)$ *is a monotone ordered resource semiring whose Bellman–Knuth dynamic program extracts minimum-risk acyclic RevIR schedules, and Grades Φ_1, Φ_2, and Φ_3 are strictly pairwise independent (Table 2).*

Table 3. QRAM-free compilation of high-level control and memory operations into $\mathcal{G}_{\text{phys}}$.

High-Level Operation	Realization in $\mathcal{G}_{\text{phys}}$	Exact Gate and Ancilla Cost
Early exit (`if |c> break`)	Nested Grover / VTAA [5]	$k_1 = 4$ fast prefix reflections (800 vs. 3,072 Toff.)
Classical table $T[\|i\rangle]$	Select-SWAP (QROAM) [9, 75]	$\lceil M/B \rceil + 2Bw$ Toffolis, Bw buffer qubits
Quantum array $\|A[\|i\rangle]\rangle$	Bitonic [14] or Bentley–Saxe [13]	$\frac{1}{4}\log_2 M(\log_2 M + 1)\,w$ (amort.) or $O(w \log^2 M)$
Variable shift $\|x\rangle \gg \|k\rangle$	$\lceil \log_2 N \rceil$-level Fredkin shifter	$N\lceil \log_2 N \rceil$ CSWAPs (0 gates for const. shifts)
Modular inverse $\|a^{-1} \bmod p\rangle$	$2N$-step `divsteps` [15] + pebbles	Fixed $2N$ steps, $16s$ pebble qubits (64 Q at $s = 4$)
Phase rotation $R_z(\theta_k)$	Phase-gradient catalyst [38]	$b - 1 = 15$ Toff. ($b = 16$, 0 online `gridsynth` Ts)

3.4 Continuous Block-Encoded Linear Algebra (QBlock)

Quantum singular value transformation (QSVT) and linear combinations of unitaries (LCU) [42, 69, 74] embed a non-unitary matrix A/α into the $|0\rangle^{\otimes m_{\text{anc}}}$ block of a unitary U. Composing block-encodings naively creates three bugs: (1) allocating fresh block ancillas at each Chebyshev step inflates width to $d \cdot m_{\text{anc}}$; (2) applying the degree-2 Chebyshev walk shortcut $W^2 = T_2(A/\alpha)$ when $U^2 \neq I$ corrupts the operator spectrum; and (3) leaving residual TaintedX phases inside PREP† alters relative LCU branch signs and cancels the target Hamiltonian sum.

LakefillQ types block-encodings via $\mathsf{QBlock}(A, \alpha, m_{\text{anc}}, \varepsilon, \Phi_X)$ inside the same judgment (Rule T-QSVTFused in Figure 2). Enforcing $\Phi_X = 0$ on PREP† unlocks Berry–Gidney MBU uncomputation [16], cutting $L = 2{,}048$ FeMoCo tensor-hypercontracted (THC) PREP† cost from 2,046 down to 91 TempANDs (22.48× reduction), while verifying $U^2 = I$ fuses Chebyshev reflections across degree d from $d \cdot m_{\text{anc}}$ down to $m_{\text{anc}} + 1$ ancillas (192 → 13 ancillas at $d = 16$).

Theorem 3.4 (QBlock Soundness and Ancilla Fusion)**.** *For even d, if $\Gamma \vdash U : \mathsf{QBlock}(A, \alpha, m_{\text{anc}}, \varepsilon_0, 0) \dashv \Gamma$ obeys Laws U1–U4 and $U^2 = I$, then* `(qsvt-cheb d U)` *implements $T_d(A/\alpha)$ within spectral error $4d\varepsilon_0$ on $m_{\text{anc}} + 1$ block ancillas instead of $d \cdot m_{\text{anc}}$.*

4 PURELY FUNCTIONAL QUANTUM DATA STRUCTURES IN LAKEFILLQ

Why do purely functional data structures matter for quantum compilation? In classical functional programming [84], persistent data structures avoid copying an N-element container by isolating updates to an $O(1)$ focus or an $O(\log N)$ structural spine. In fault-tolerant quantum circuits, two physical uncomputation laws turn this exact structural locality into large reductions in logical qubits, Toffoli count, and QEC code distance.

4.1 Physical Uncomputation Boundaries Motivating Functional Locality

All data structures in LakefillQ lower to $\mathcal{G}_{\text{phys}}$ without QRAM (Table 3). Two sharp algebraic boundaries determine when temporary ancillas can be erased for free and how many registers tree-structured computations require.

Proposition 4.1 (Exact Algebraic Degree-2 Boundary of Clifford MBU)**.** *Measuring an ancilla $|f(x)\rangle$ ($x \in \{0,1\}^n$) in the X basis yields outcome $m \in \{0,1\}$ and residual state $\sum_x \alpha_x (-1)^{m \cdot f(x)} |x\rangle$. The phase $(-1)^{m \cdot f(x)}$ can be canceled across all x using exclusively Clifford gates (Z, S, CZ, zero T or Toffoli gates)* if and only if *$\deg_{\mathbb{F}_2}(f) \leq 2$. For any Boolean function with $\deg_{\mathbb{F}_2}(f) \geq 3$ (such as $x_1x_2x_3$), Reed–Muller distance to $\mathcal{RM}(2, n)$ caps the worst-case state fidelity of any Clifford fixup on the uniform superposition at $\leq 9/16 = 0.5625$.*

Proposition 4.2 (Sequential Two-Pass vs. Simultaneous Tree Pebbling)**.** *Evaluating a height-h binary Merkle or sponge tree of W-qubit digests by keeping both child digests live simultaneously requires $(2h+1)W$ qubits. Factoring parent compression into two sequential child passes (S_{left} followed*

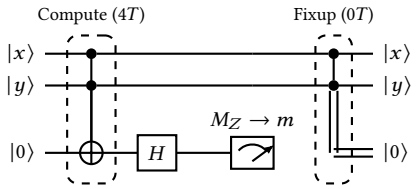


Fig. 3. Gidney's degree-2 temporary-AND compute/uncompute elbow [38]: computing $|x \wedge y\rangle$ costs $4T$ (1 Toffoli), whereas X-basis measurement plus classical feed-forward CZ^m erases the ancilla with $0T$ because $\deg_{\mathbb{F}_2}(x \wedge y) = 2$.

Table 4. The nine core purely functional quantum data structures in LakefillQ, mapping classical persistence ideas to quantum hazards eliminated and mechanized resource bounds.

Data Structure	Classical Origin	Hazard Eliminated	Mechanized Payoff
`HuetFocusZipper`	Huet zipper [56]	Cold-limb aliasing	0 cold Toff.; w-bit focus
`SkewBinaryCarrySave`	Skew-binary [82]	C-limb carry (Δd=+2)	$O(1)$ carry; keeps $d = 27$
`BraunTree`	Braun tree [21]	Heap pointers & tags	0 ptr Q; $\lceil \log_2 k \rceil$ depth
`PackedBezoutZipper`	Ext.-GCD [15]	$4N$-bit (u, v, r, s) bloat	$2N$ Q (−512 Q ECC-256)
`BinomialPebble`	Pebbling [12, 84]	Deg-> 2 history tape	$\binom{s+d}{s}$ steps in sW Q
`MonoidalFingerTree`	Finger tree [52]	Full-width IP checks	Prefix VTAA (800 vs. 3,072)
`StreamingSyndrome`	Pivoted Bruhat LU	Dense r^2 parity matrix	r-bit (76.3×–4,107×)
`CritBitBitonicDict`	Crit-bit [13, 81]	History garbage bits	0 hist. Q; 74.5× amort.
`WalkerVoseAlias`	Walker–Vose [109, 110]	Deep arccos trees	$O(1)$ depth; 280 Toff. (7.31×)

by S_{right}, uncomputing the left child before evaluating the right) reuses a single h-register child pool, achieving the Sethi–Ullman bound of $(h+1)W$ qubits [104] and saving hW qubits without increasing depth order.

The Unifying Quantum Data-Structure Principle. Propositions 4.1 and 4.2 explain *why* imperative flat arrays struggle on fault-tolerant hardware and *how* functional data structures rescue them:

(1) **Confine non-Clifford work to a narrow focus** where intermediate `TempANDs` remain quadratic ($\deg_{\mathbb{F}_2} \leq 2$) in their local inputs and uncompute immediately via zero-Toffoli MBU (§4.2).
(2) **Bound high-degree (**$\deg > 2$**) multi-step trajectories**—where Proposition 4.1 forbids free Clifford MBU—using complementary-length state packing and Královič–Bennett binomial checkpoints [12, 67] (§4.3).
(3) **Eliminate structural garbage bits** using pointerless implicit trees, monoidal prefix summaries, streaming accumulators, and history-independent canonical dictionaries (§4.4).

Table 4 summarizes all nine core data structures in LakefillQ, showing the classical inspiration, the quantum hazard that breaks a naive translation, and the resulting circuit guarantee.

4.2 Family I: Focus Zippers, Skew-Binary Carry-Save Numerals, and Braun Trees

1. Huet Focus Zipper (`HuetFocusZipper`*).* Classically, Huet's zipper [56] represents a sequence with a cursor as a triple (Left, Focus, Right) so local edits touch only Focus in $O(1)$ time. In a multi-precision quantum circuit over C limbs of w qubits ($N = Cw$), wiring arithmetic gates across all N qubits simultaneously forces dense long-range routing and exposes idle limbs to accidental control/target aliasing. In `#lang lakefillq`, `focus-zipper` partitions an N-qubit register into one w-qubit mutable focus (Lin1) and $(C-1)w$ cold context qubits (BorrowZ). Advancing the focus

by a compile-time constant limb offset costs *zero quantum gates* (pure compiler wire relabeling), while quantum-controlled shifts use a $\lceil \log_2 C \rceil$-level Fredkin network.

Proposition 4.3 (`HuetFocusZipper` Locality)**.** *A C-limb, w-bit* `HuetFocusZipper` *executes focus-local operations with zero Toffoli gates on the $(C-1)w$ cold context qubits, zero gates for static cyclic shifts, and $+2\lceil\sqrt{Q_{\text{cold}}}\rceil$ planar surface-code bus tiles separating hot magic-state factories from cold memory.*

2. Skew-Binary Carry-Save Numeral (`SkewBinaryCarrySaveNumeral`*).* Streaming C limbs through a single w-bit zipper focus creates a subtle fault-tolerant trap: if adding limb k must wait for the carry out of limb $k-1$, C limbs execute strictly sequentially, stretching circuit depth by $C\times$ and forcing QEC code distance $d \mapsto d+2$ on RSA-2048 (§5). Myers' classical skew-binary numeral system [82, 84] eliminates carry cascades by allowing at most one sparse digit 2 so increments run in $O(1)$ worst-case steps. Quantumly, `SkewBinaryCarrySaveNumeral` augments each w-bit limb boundary with a 2-qubit redundant carry rail $d_k \in \{0,1,2\}$. Each modular addition absorbs carries locally in $O(1)$ limb depth without rippling across the remaining $C-1$ limbs; only once, immediately prior to QFT or final comparison, does a single canonical ripple pass discharge the redundant rails to satisfy Law U2 (Listing 4 and Figure 4).

Proposition 4.4 (`SkewBinaryCarrySaveNumeral` Carry Decoupling and Canonical Collapse)**.** *Each window addition on a C-limb* `SkewBinaryCarrySaveNumeral` *completes in $O(1)$ limb depth using $2C$ redundant boundary qubits, and a single terminal ripple pass bijectively restores unique binary canonicality (*U2*) while preserving exact integer value modulo 2^N.*

3. Implicit Pointerless Braun Tree (`BraunTree`*).* Classical heap and tree reductions often store explicit child pointers or AVL balance tags—bits that become entangled quantum garbage when superposed branches rebalance differently. A Braun tree [21, 83] enforces the size invariant $0 \le |L| - |R| \le 1$ at every node, determining routing solely from the least-significant bit of the index with *zero pointer or balance qubits*. In `#lang lakefillq`, `(fold-braun op xs)` lowers associative reductions (popcount, RNS-CRT product trees, syndrome parity folds) into an exact $\lceil \log_2 k \rceil$-depth balanced tree.

Proposition 4.5 (`BraunTree` Pointerless Logarithmic Reduction)**.** *For k leaves of width w,* `fold-braun` *allocates zero pointer or balance-tag qubits, completes in exact tree depth $\lceil \log_2 k \rceil$, and consumes $(k-1)w$ Toffoli gates under MBU cleanup.*

Listing 4. Composing `within-apply-mbu`, `focus-zipper`, and skew-binary local carry folding in `#lang lakefillq`.

```
;; O(1)-depth local skew-binary carry absorption on active limb focus `limb-k` + 2-bit rail `rail-k`
(defqfree (skew-window-add #:wires ([addend (range 0 64)] [limb-k (range 64 128)] [rail-k (range 128 130)])
                           #:scratch ([c64 (range 130 194)]))
  (within-apply-mbu
   (gidney-carry-chain addend limb-k c64)
   (focus-zipper [fz (append limb-k rail-k) (append addend limb-k rail-k)]
     (seq (gidney-sum-xor addend fz c64)
          ;; Absorb carry-out c64[63] locally into 2-bit skew rail {0,1,2} without C-limb ripple:
          (ccx (wref c64 63) (wref fz 64) (wref fz 65))
          (cx  (wref c64 63) (wref fz 64))))
   c64))
```

4.3 Family II: Packed Bézout Shift-Zippers and Binomial Pebble Schedules

4. Packed Bézout Shift-Zipper (`PackedBezoutShiftZipper`*).* Elliptic-curve discrete-logarithm (ECDLP) circuits spend most of their width and gates on modular inversion $|a^{-1} \bmod p\rangle$ via

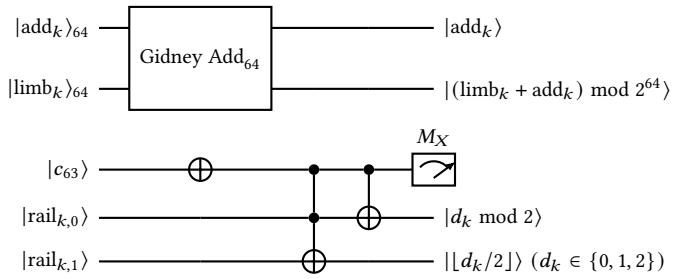


Fig. 4. Local carry absorption in `SkewBinaryCarrySaveNumeral`: carry-out $|c_{63}\rangle$ folds into the 2-qubit rail $|\text{rail}_k\rangle$ in depth 2, decoupling all C limbs during modular exponentiation before a single pre-QFT canonical collapse.

constant-time extended binary GCD (`divsteps`) [15, 48, 98], which tracks four N-bit integers: remainders (f, g) and Bézout cofactors (u, v) (or (r, s)). Naively storing all four N-bit registers requires $4N$ qubits (1,024 Q at $N = 256$). Yet throughout binary GCD, the bit-lengths of u and r grow only as the bit-lengths of f and g shrink, obeying $\text{len}(u) + \text{len}(f) \leq N + 1$ and $\text{len}(v) + \text{len}(g) \leq N + 1$. `PackedBezoutShiftZipper` packs (u, f) and (v, g) head-to-tail into *two* N-qubit double-ended shift-zippers around a $2J$-bit active LSB focus ($J = 16$), saving $2N = 512$ logical qubits on ECC-256. Furthermore, to compress modular multiplication ($x \cdot y \bmod p$) inside point addition, we apply Hua's identity $xy = \binom{x+y}{2} - \binom{x}{2} - \binom{y}{2}$ [54] with explicit singular $\{0, 1\}$ endpoint corrections so Law U1 holds across all residues in $\mathbb{F}_p$.

Proposition 4.6 (`PackedBezoutShiftZipper` $2N$-Qubit Packing and Totalized Hua Identity)**.** *Packing complementary-length pairs (u, f) and (v, g) around a $2J$-bit focus reduces GCD state width from $4N$ to $2N$ qubits with exact half-domain cofactor bijection on $\mathbb{F}_p$, and totalized Hua squaring preserves bijective reversibility across all singular inputs $x \in \{0, 1\}$.*

5. Binomial Pebble Schedule (`BinomialPebbleSchedule`*).* Across $2N = 512$ `divsteps`, the cumulative branch trajectory has algebraic degree $\deg_{\mathbb{F}_2} \gg 2$, so Proposition 4.1 forbids erasing the 512-step history via Clifford MBU alone, while storing a 1-bit swap flag per step costs 512 history qubits. Grouping $J = 16$ steps into macro-windows yields $L = 32$ irreversible steps, which Královič–Bennett binomial pebbling [12, 67] spans using $s = 4$ pebbles of width $W = 16$ (64 pebble qubits total, since $\binom{4+4}{4} = 70 \geq 32$) in 109 macro-evaluations instead of a 512-qubit history tape.

Proposition 4.7 (`BinomialPebbleSchedule` Space–Time Optimality)**.** *With s checkpoint pebbles of width W at recursion depth d,* `BinomialPebbleSchedule` *spans up to $\binom{s+d}{s}$ high-degree ($\deg_{\mathbb{F}_2} > 2$) irreversible windows in sW pebble qubits (64 Q at $s = 4, W = 16$) and $\sum_{k=1}^{d} 2^k \binom{s+d-k-1}{s-1}$ window evaluations (109 for $L = 32$), returning all pebbles to $|0\rangle$.*

Listing 5. Jump-16 packed Bézout step in `#lang lakefillq`: non-Clifford `divsteps` touch only the $2J$-bit focus while binomial pebbles checkpoint the 16-bit swap trajectory.

```
(defq (packed-bezout-jump16 #:wires ([uf-zip (range 0 256)] [vg-zip (range 256 512)]
                                     [pebble-k (range 512 528)])
                            #:scratch ([t-mat (range 528 560)])
                            #:domain (domain-guard odd-modulus-nonzero-guard))
  ;; 1. Compute 2x2 Jump-16 transition matrix and 16-bit swap history on the 32-bit LSB focus:
  (focus-zipper [lsb-focus (append (take uf-zip 32) (take vg-zip 32) pebble-k)
                           (append uf-zip vg-zip pebble-k)]
    (within-apply-mbu
     (jump16-lsb-divsteps lsb-focus t-mat)
```

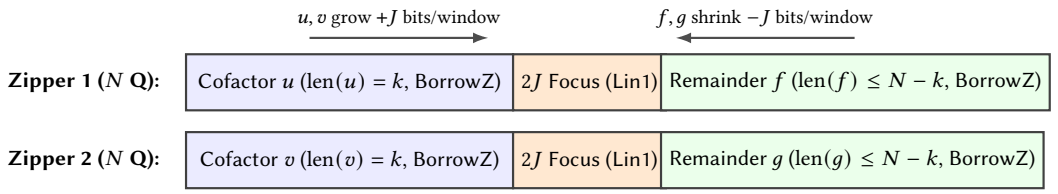


Fig. 5. Complementary-length packing in `PackedBezoutShiftZipper`: because $\text{len}(u) + \text{len}(f) \leq N + 1$ throughout binary GCD, (u, f) and (v, g) share two N-qubit registers around a $2J$-bit active LSB focus rather than occupying four N-qubit registers (saving 512 Q at $N = 256$).

```
    (record-pebble-history t-mat pebble-k)
    t-mat))
;; 2. Apply linear cofactor/remainder combination across full zipper & shift boundary by 16
(apply-bezout-transition-and-shift16 uf-zip vg-zip pebble-k))
```

4.4 Family III: Monoidal Filters, Streaming Matrices, Canonical Tries, and Alias Tables

6. Monoidal Finger-Tree Filter (`MonoidalFingerTreeFilter`*).* Hinze and Paterson's finger trees [52] annotate internal spine nodes with monoidal summaries so searches prune subtrees before inspecting leaves. In quantum search (such as SimHash inner-product filtering in lattice sieving or substring matching), computing full high-precision inner products on every candidate wastes thousands of Toffolis on non-matching states. `MonoidalFingerTreeFilter` caches PopCount/XOR prefix summaries and pairs them with Two-Level VTAA [5], rejecting non-matching candidates in 255 forward Toffolis and lowering average reflection cost from 3,072 to 800 Toffolis/step.

Proposition 4.8 (`MonoidalFingerTreeFilter` Prefix Pruning)**.** *Caching associative monoidal prefix measures reduces fast-reject reflection cost to* 255 *forward Toffolis and cuts amortized VTAA step cost from* 3,072 *to* 800 *Toffolis (*3.84×*).*

7. Bruhat Pivoted LU and Streaming Syndrome Accumulator (`StreamingSyndromeAccumulator`*).* Information-Set Decoding (ISD) for code-based cryptography (Classic McEliece, BIKE) traditionally allocates an entire $r \times r$ binary parity-check submatrix on quantum wires ($r^2 = 590{,}000+$ qubits for McEliece-348864) to perform Gaussian elimination in superposition [34, 58]. Instead, for small dense blocks ($r \leq 64$), `BruhatPivotedLU` factors nonsingular matrices in $\text{GL}_r(\mathbb{F}_2)$ in place using $r\lceil \log_2 r \rceil$ canonical pivot qubits; for large code-based instances, `StreamingSyndromeAccumulator` keeps only an r-qubit syndrome register live and streams classical columns controlled on low-weight error positions, shrinking McEliece-348864 width by 76.26× (to 7,744 Q) and BIKE-L1 width by 4,107× (to 36,969 Q).

Proposition 4.9 (`BruhatPivotedLU` and `StreamingSyndromeAccumulator` Width Compression)**.** `BruhatPivotedLU` *factors every* $A \in \text{GL}_r(\mathbb{F}_2)$ *bijectively with* $r\lceil \log_2 r \rceil$ *pivot bits, while* `StreamingSyndromeAccumulator` *eliminates* $O(r^2)$ *matrix allocation in favor of an* $O(r)$*-qubit accumulator.*

8. Canonical Crit-Bit and Bitonic Dictionary (`CanonicalCritBitBitonicDictionary`*).* In quantum walk algorithms (element distinctness [4], subset sum, Kuperberg's collimation sieve), two paths reaching the same subset S must interfere constructively: $|\text{Dict}(S)\rangle$ must depend *solely* on the set S, never on insertion order. Balanced AVL or red-black trees violate Law U2 because tree

shape depends on insertion history; pointer-based radix trees [114] require dynamic freelist management. `CanonicalCritBitBitonicDictionary` combines Morrison's history-independent crit-bit splits [81] with Batcher bitonic sorting networks [14] (55 comparator layers for $M = 1{,}024$) and Bentley–Saxe logarithmic level merges [13], storing zero rotation or pointer garbage bits and amortizing dictionary maintenance by 74.47× across parallel quantum walkers.

Proposition 4.10 (`CanonicalCritBitBitonicDictionary` History Independence)**.** *Every M-element set S has a unique bit representation independent of insertion permutation (unit overlap $\langle \mathrm{Dict}_{\pi_1}(S) \mid \mathrm{Dict}_{\pi_2}(S) \rangle = 1$, satisfying Law* U2*), with* 74.47× *per-walker Toffoli amortization under Batcher bitonic batching.*

9. Walker–Vose Coherent Alias Table (`WalkerVoseAliasTable`*).* Preparing coherent state superpositions $\sum_{i=0}^{2^w-1} \sqrt{p_i}|i\rangle$ for LCU/QSVT chemistry Hamiltonians via recursive arccos rotation trees requires $O(2^w)$ sequential controlled rotations. Adapting Walker–Vose alias sampling [9, 109, 110], `WalkerVoseAliasTable` prepares uniform index and coin registers, looks up (q_i, alias_i) via Select-SWAP, and performs a single μ-bit reversible comparison + Fredkin swap in $O(1)$ comparator depth ($2w + 2\mu + 1 = 49$ Q, $2^w + \mu + w = 280$ Toffolis at $w = 8, \mu = 16$, 7.31× fewer Toffolis than arccos trees; Listing 6). Four bootstrapped containers—`NestedDyadicWalkerAliasBox`, `BrodalOkasakiBootstrappedSkewHeap`, `CombinadicsSkewRALSimplex`, and `OkasakiBinomialCollimationQueue`—compose these nine primitives across 60 compiled workloads (Appendix C).

Listing 6. Coherent Walker–Vose alias state preparation in `#lang lakefillq`: $O(1)$-comparator Select-SWAP + Fredkin swap with Law-U4 MBU scratch erasure.

```
(defq (walker-vose-alias-prep #:wires ([idx (range 0 8)] [coin (range 8 24)]
                                       [keep-q (range 24 40)] [alias-idx (range 40 48)]
                                       [cmp-flag (list 48)])
                              #:scratch ([cmp-c (range 49 65)]))
  (for/seq ([b (in-range 8)])  (h (wref idx b)))
  (for/seq ([c (in-range 16)]) (h (wref coin c)))
  (select-swap-qroam alias-table idx (append keep-q alias-idx))
  (within-apply-mbu
   (gidney-lt-carry coin keep-q cmp-c)
   (cx (wref cmp-c 15) (wref cmp-flag 0))
   cmp-c)
  ;; Swap `idx` with `alias-idx` when coin >= keep-q (cmp-flag == 0):
  (x (wref cmp-flag 0))
  (for/seq ([b (in-range 8)]) (cswap (wref cmp-flag 0) (wref idx b) (wref alias-idx b))))
```

Proposition 4.11 (`WalkerVoseAliasTable` $O(1)$-Comparator State Preparation)**.** *For 2^w probabilities discretized to μ bits,* `WalkerVoseAliasTable` *prepares $\sum_i \sqrt{p_i}|i\rangle$ (with clean entangled coin ancillas handled by* QBlock *in LCU) using $2w + 2\mu + 1$ qubits and $2^w + \mu + w$ Toffolis.*

5 HARDWARE-GRADED LOWERING, ZX E-GRAPH SATURATION, AND PARTITIONING

Once $\lambda_{\mathrm{LFQ}}^{\mathrm{kern}}$ certifies logical purity (U1–U4), the fastest and highest-fidelity circuit still depends dramatically on the target architecture: mid-circuit MBU excels when an external QEC syndrome decoder keeps up ($\tau_{\mathrm{dec}} \leq \tau_{\mathrm{cycle}}$), whereas high readout noise favors measurement-free Margolus `RCCX` [78] or native Rydberg `CCZ` [18]. LakefillQ resolves these trade-offs automatically in RevIR before emitting single- or multi-QPU MachineISA.

Table 5. Bellman–Knuth QPhysGrade e-graph extraction across four hardware profiles, comparing baseline reversible extraction against non-Clifford ZX phase-polynomial saturation on ideal and noisy density-matrix simulators.

Target Profile	Optimal Gadget	Base Surv.	+ZX Surv.	Gain	Ideal
Transmon NISQ ($\epsilon_{\text{meas}} > 2\epsilon_{2Q}$)	Meas.-free RCCX + ZX	0.9286 (0.8273 raw)	0.9611	+3.25%	1.0000
IBM Heron r2 ($\tau_{\text{dec}} = 420$ ns)	MBU+XY4 + ZX (d=5: 0.9973)	0.9101 (0.8175 raw)	0.9524	+4.23%	1.0000
QuEra Bluvstein (90% erasure)	Native RydbergCCZ + ZX	0.9718 (0.9828 1-blk)	0.9937	+2.19%	1.0000
FTQC BB-qLDPC ($\tau_{\text{dec}} = 340$ ns)	MBU+QROAM + Twin-CCZ ($8T$)	0.9966 (0.9903 raw)	0.9984	+0.19%	1.0000

5.1 Non-Clifford ZX Phase-Polynomial Saturation and Four-Target Extraction

After asymmetric TempAND/MBU lowering and chordal SSA wire coloring, LakefillQ populates a congruence-closed RevIR e-graph combining reversible Boolean identities, target gadget decompositions (RCCX, MBU+XY4, RydbergCCZ, MBU+QROAM), and two non-Clifford ZX phase-polynomial rewrite families [31, 65]:

(1) **ZX Parity-Phase Spider Fusion.** Adjacent CNOT-sandwiched phase gadgets $\mathrm{CX}(a,b) \cdot T^{\pm k}(b) \cdot \mathrm{CX}(a,b)$ lift into canonical parity-phase spiders $e^{i(\pm k\pi/4)(x_a \oplus x_b)}$ whose angles add modulo 8; opposing T and $T^\dagger$ spiders across commuting CNOTs fuse and cancel ($2T \to 0T$).
(2) **Shared-Control Twin-CCZ Cascade.** Two CCZ or TempAND elbows sharing controls (c_0, c_1) on targets (t_0, t_1) rewrite from two independent $8T$ Clifford+T expansions ($16T$ total) into a fused ZX phase-polynomial gadget requiring $8T$ total (50% T-count reduction) while preserving exact Boolean and phase semantics.

Bellman–Knuth dynamic programming over QPhysGrade extracts the minimum-risk schedule per hardware profile (Table 5). Across our multi-gadget benchmark (31 e-classes, 81 e-nodes, 60 equivalence merges), ZX spider fusion improves noisy density-matrix survival across every target (+3.25% on NISQ, +4.23% on IBM Heron r2 [57], +2.19% on QuEra Bluvstein [18, 19], and +0.19% on BB-qLDPC [22]). When Φ_1 is instantiated with external decoder latency τ_{dec}, fast cryo-CMOS ($\tau_{\text{dec}} = 120$ ns) and 300 K coax FPGA Union-Find ($\tau_{\text{dec}} = 420$ ns) both extract the $32T$ MBU+ZX schedule (60,960 ns and 63,360 ns), whereas an unpaced software MWPM decoder ($\tau_{\text{dec}} = 4{,}500$ ns) automatically triggers fallback to the measurement-free $64T$ unitaryZX schedule (124,000 ns) rather than suffering Lindley queue divergence. Whenever a user program violates Φ_1–Φ_3, repairPhysGrade synthesizes counterexample-guided repairs (XY4 dynamical decoupling, exclusion coloring, flag-ancilla insertion, or MBU-to-RCCX fallback).

Lemma 5.1 (Bernoulli Survival Monotonicity)**.** *Under independent gate depolarization rates $p_i \in [0, 1)$, log-survival $-\sum_i \ln(1 - p_i)$ is strictly additive under ;; and monotone with respect to* QPhysGrade.

Lemma 5.2 (Heralded Erasure Fault-Distance Doubling)**.** *On neutral-atom architectures with* 90% *heralded Rydberg erasure conversion [18, 43], t_{Pauli} unheralded faults and t_{erase} heralded erasures are correctable at odd distance d whenever $2t_{\text{Pauli}} + t_{\text{erase}} \le d - 1$, doubling heralded fault tolerance relative to unheralded Pauli errors and reducing logical error rates by* 6.5×–36×.

5.2 Code-Distance Step-Down, qLDPC Automorphisms, and Leaf Synthesis

Why Carry Decoupling Saves Physical Qubits. In fault-tolerant surface codes [40, 72], logical error per round scales as $p_L(d) \approx c(p/p_{\text{th}})^{(d+1)/2}$, so total spacetime volume $V = Q_{\log} \cdot D_{\text{cycles}}$ must satisfy $V \cdot p_L(d) \le \delta_{\text{target}}$. If compressing logical width $Q_{\log}$ inflates depth D_{cycles} by 32× (as happens when 32 RSA-2048 limbs ripple carries sequentially through one 64-bit focus), V rises

past the $d = 27$ budget and forces $d = 29$—increasing physical qubits per tile by $(29/27)^2 \approx 15.4\%$. Theorem 5.3 formalizes the converse payoff achieved by `SkewBinaryCarrySaveNumeral` and Early-MBU compression.

Theorem 5.3 (Spacetime Volume to Code-Distance Step-Down)**.** *At physical error rate $p < p_{\text{th}}$, reducing active logical spacetime volume from V_0 to $V_1 \leq V_0 \cdot (p/p_{\text{th}})^k$ reduces the minimum odd QEC code distance required for target failure budget δ by $\Delta d \geq 2k$, while skew-binary limb decoupling preserves $d = 27$ (rather than $d = 29$) on RSA-2048 (*891,348 *yoked surface-code or* 344,304 *BB-qLDPC physical qubits).*

Corollary 5.4 (Cat-Qubit Bias Preservation and Photonic Distillation)**.** *Restricting focus gates to Z-bias-preserving operations (*`CX`*,* `CCZ`*, M_X) preserves Kerr-cat bit-flip suppression (*76× *physical overhead reduction), and double-ended photonic EPR zipper buffering with DEJMPS distillation [30, 96] maintains inter-module threshold soundness.*

Cold-Context qLDPC Automorphisms and Algorithmic Lowerings. Because > 80% of logical qubits in zipper-structured programs reside in passive cold limbs (`BorrowZ`), MachineISA maps cold context onto high-rate yoked surface codes [39] or [[288, 12, 18]] Bivariate Bicycle qLDPC blocks [22], where cyclic limb shifts execute in hardware depth 1 without ancilla bridges (Proposition 5.5), and onto neutral-atom AOD arrays where skew-binary carry decoupling enables 37.8× faster single-move parallel limb shuttling. At the leaf level, a 64-bit bitboard Meet-in-the-Middle (MITM) superoptimizer [6] compresses 4-to-5-qubit arithmetic and S-box subcircuits by 41.9% alongside four compiler-derived cryptographic syntheses (Theorem 5.6).

Proposition 5.5 (Bivariate Bicycle qLDPC Torus Shift Automorphism)**.** *On any Bivariate Bicycle code over $\mathbb{F}_2[x, y]/(x^\ell - 1, y^m - 1)$ (including [[288, 12, 18]]), every cyclic torus shift $T_{a,b} = S_x^a S_y^b \oplus S_x^a S_y^b$ (multiplying both block coordinates by $x^a y^b$) commutes with both check matrices H_X, H_Z, implementing cyclic* `HuetFocusZipper` *shifts in physical depth* 1 *with zero ancilla bridge qubits.*

Theorem 5.6 (Sublinear QROAM, Quotient RNS, AES, and Early-MBU DQI Syntheses)**.** *Within λ_{LFQ}, asymmetric MBU and* `QSub` *derive: (i)* `34-TempAND` *Boyar–Peralta [20] +* `91-CX` *Mix-Columns [112] AES rounds (*`32 TempANDs` *under* `BP32`*,* `5` *ancillas under* `DLog-5` *[70]); (ii) $O(\sqrt{N})$ Berry–Gidney Select-SWAP table erasure [16] (*61 *vs.* 254 `TempANDs` *at $N = 256$); (iii) Law-*`U2` *Kawamura fixed-point RNS-CRT quotient erasure [62]; and (iv) exact Early-MBU DQI error-register reclamation prior to syndrome decoding across Hermitian (H_2, H_3, H_4), lifted bivariate ($\mathbb{F}_4^2$), and Suzuki ($S(8)$) algebraic-geometry codes.*

5.3 Multi-QPU Partitioning, Distributed QRNS, and Law-U4 GHZ Disentangling

When a circuit exceeds single-chip capacity Q_{cap} (`lakefillqc --emit-cluster-dir`), naive gate-cut teleportation across M QPUs incurs heavy optical Bell-pair overhead from cross-chip carry chains and reverse lookup uncomputation [47, 119]. LakefillQ combines three distributed mechanisms:

(1) **Zero-Carry Distributed QRNS.** Modular exponentiation factors across coprime RNS moduli per QPU so bulk arithmetic consumes zero inter-QPU carry Bell pairs, synchronizing only at compact Braun-tree Kawamura CRT reductions.
(2) **Law-U4 GHZ Cat-Disentangling.** Instead of spending $2k$ Bell pairs to fan out a k-bit table address and reverse-teleport it back, LakefillQ measures remote address copies in the X basis (M_X) and cancels the remote phase kickbacks locally via Law-`U4` `cond-z` feed-forward—cutting optical Bell pairs by 50.0% (0 reverse ebits) and lowering Stim+PyMatching decoded logical error rates under DEJMPS distillation by 2.27×–16.00×.

(3) **Convex M^* Selection (T-ParGuard).** Over $M \in \{1, \ldots, 256\}$, Rule T-ParGuard minimizes total physical spacetime volume: on unstructured Grover/ISD search where Zalka's $\sqrt{M}$ parallel overhead [116] outweighs width splitting, it automatically selects $M^* = 1$ whenever single-chip capacity permits; on Shor/QRNS workloads, it adapts $M^*(p_{\text{link}})$ smoothly to optical link noise (contracting RSA-2048 $M^* = [6, 6, 4, 2, 1]$ as p_{link} degrades from 10^{-4} to $5{\times}10^{-2}$).

6 EVALUATION

We evaluate LakefillQ across six research questions examining algebraic exactness and non-binary AG DQI (**RQ1**), differential fuzzing (**RQ2**), mutation testing and error diagnostics (**RQ3**), mechanized metatheory (**RQ4**), empirical relevance of our typing side conditions on third-party circuits (**RQ5**), and open-problem Pareto advances across execution regimes (**RQ6**).

6.1 RQ1: Cryptographic Oracles and Non-Binary Algebraic-Geometry DQI

Compiled LakefillQ circuits match official NIST FIPS-197 test vectors bit-for-bit across AES-128 (`69c4e0d86a7b0430d8cdb78070b4c55a`), AES-192 (`dda97ca4864cdfe06eaf70a0ec0d7191`), and AES-256 (`8ea2b7ca516745bfeafc49904b496089`), match NIST FIPS-202 Keccak-$f[200]$ permutations, and pass exhaustive truth-table verification on BCH$(15, 7, 5)$, $\mathbb{F}^*_{65521}$ binary GCD, $\mathrm{GL}_4(\mathbb{F}_2)$ Bruhat LU across all 20,160 nonsingular matrices, $\mathbb{Z}^2_{3329}$ odd-half NTT butterflies, and [[288, 12, 18]] torus automorphisms.

Table 6 evaluates our Early-MBU DQI compiler pass across non-binary Hermitian curves [35, 105], lifted bivariate affine evaluation codes, and Deligne–Lusztig Suzuki curves over $\mathbb{F}_q$, comparing against unitary syndrome-decoding DQI [60]. On $[8, 5, 3]_{\mathbb{F}_4}$ ($H_2 : x^3 = y^2 + y$, whose blocklength $n = 8$ strictly exceeds the $\mathbb{F}_4$ Reed–Solomon MDS ceiling $q + 1 = 5$), `early_mbu_dqi_hermitian_ag_f4.lfq` compiles with zero flags (46 virtual SSA wires chordally colored onto 22 physical qubits, shrinking post-encoding live logical width from 22 to 6 qubits and cutting T-gates from 336 to 96 with 0 reverse Toffolis), achieving statevector fidelity 1.000000 against unitary DQI across all 25 coset leaders and all 32 M_X trajectories (peak interference 0.108114 at syndrome 57). On the lifted bivariate evaluation code $[9, 6, 3]_{\mathbb{F}_4}$ over $\{0, 1, \omega\}^2 \subset \mathbb{F}_4^2$ (`early_mbu_dqi_lifted_bivariate_f4.lfq`, $n = 9 > 5$), 51 virtual SSA wires color onto 24 physical qubits and Early-MBU shrinks live width by 4.00× (24 → 6 Q, 27 `TempANDs`, 504 → 108 T, 1.000000 fidelity across 28 coset leaders). Scaling to higher-genus curves compresses live logical width by 5.50× on genus-3 $H_3/\mathbb{F}_9$ ($[27, 21, 4]_{\mathbb{F}_9}$, 132 → 24 Q, 189 `TempANDs`), 6.33× on the genus-14 Suzuki curve $S(8) : y^8 + y = x^2(x^8 + x)$ over $\mathbb{F}_8$ ($[64, 52, 6]_{\mathbb{F}_8}$, where $n = q^2 = 64 \gg n_{\text{RS}} \leq 9$, 228 → 36 Q, 320 `TempANDs`, 5,824 → 1,280 T), 6.33×–6.82× on genus-6 $H_4/\mathbb{F}_{16}$ (304 → 48 Q and 3.18× T-gate gain on $[64, 52, 7]_{\mathbb{F}_{16}}$; 300 → 44 Q and 8,064 → 1,792 T, 4.50×, on $[64, 53, 6]_{\mathbb{F}_{16}}$), and 9.00× on $[512, 448, 37]_{\mathbb{F}_{64}}$ (3,456 → 384 Q).

6.2 RQ2: Six-Engine Differential Fuzzing

To test end-to-end compilation thoroughly, we cross-checked LakefillQ across six independent engines with zero discrepancies: (1) 500/500 randomized density-matrix equivalence trials in Qiskit `AerSimulator`; (2) 640/640 generated `#lang lakefillq` surface programs and 2,860,800 kernel reduction steps in PLT Redex; (3) 38/38 tensor-network checks in PyZX [31, 65]; (4) 360/360 fault-injection and optical-distillation runs in Stim + PyMatching; (5) 80,000 QEC rounds of real-time decoder queue verification; and (6) Z3 SMT equivalence miters across lowered RevIR blocks.

Table 6. Non-binary Hermitian, lifted bivariate, and Suzuki algebraic-geometry Early-MBU DQI vs. unitary DQI across $\mathbb{F}_4$, $\mathbb{F}_8$, $\mathbb{F}_9$, $\mathbb{F}_{16}$, and $\mathbb{F}_{64}$ (all exceeding the univariate Reed–Solomon MDS ceiling $n_{\mathrm{RS}} \leq |\mathbb{F}| + 1$, with 1.000000 simulator fidelity on H_2, lifted $\mathbb{F}_4^2$, H_3, $S(8)$, and H_4).

AG / Lifted Code (Genus g)	RS Ceiling	Unitary Q	Early-MBU Q	Width Gain	T-Gate Gain
H_2: $[8,5,3]_{\mathbb{F}_4}$ ($g=1$)	$n_{\mathrm{RS}} \leq 5$	22	6 (9 active)	3.67×	336 → 96 (3.50×)
Lifted $\mathbb{F}_4^2$: $[9,6,3]_{\mathbb{F}_4}$ ($g=0$)	$n_{\mathrm{RS}} \leq 5$	24	6 (9 active)	4.00×	504 → 108 (4.67×)
H_3: $[27,21,4]_{\mathbb{F}_9}$ ($g=3$)	$n_{\mathrm{RS}} \leq 10$	132	24 (28 active)	5.50×	3,402 → 756 (4.50×)
Suzuki $S(8)$: $[64,52,6]_{\mathbb{F}_8}$ ($g=14$)	$n_{\mathrm{RS}} \leq 9$	228	36 (42 active)	6.33×	5,824 → 1,280 (4.55×)
H_4: $[64,52,7]_{\mathbb{F}_{16}}$ ($g=6$)	$n_{\mathrm{RS}} \leq 17$	304	48 (44 on d=6)	6.33×	8,064 → 1,792 (4.50×)
H_8: $[512,448,37]_{\mathbb{F}_{64}}$ ($g=28$)	$n_{\mathrm{RS}} \leq 65$	3,456	384	9.00×	3.12×

6.3 RQ3: Mutation Testing and Three-Span Resugared Diagnostics

Injecting 12 targeted soundness mutations into `lakefillqc` (individually disabling U1 domain restriction, U2 canonical discharge, U3 branch orthogonality, U4 M_X phase-taint tracking, chordal live-range interference, Φ_1 backlog checking, Φ_2 exclusion coloring, Φ_3 hook-flag checking, QBlock $U^2 = I$ checks, and T-ParGuard Zalka penalties; Table 13 in Appendix D) yields a 12/12 (100.0%) kill rate, with every mutant caught by concrete simulator counterexamples.

Because multipass elaboration and chordal SSA wire reuse rewrite internal wire indices, naive compiler errors blame mangled compiler temporaries [29, 63, 94]. `lakefillqc --diagnostics-human` derives three-span provenance (`origin_span`, `mutating_span`, `failure_span`) [50, 76, 90, 101, 118] directly from $\Gamma_{\mathrm{in}} \vdash t : Q_e(A) \dashv \Gamma_{\mathrm{out}}$, resugaring chordally recycled physical wires back to user bindings (e.g., `s2[0] (recycled physical wire q[12] from s1[0])`) and emitting a minimal basis ket counterexample (such as $|a_0 = 1, b_0 = 1\rangle$). Across 8/8 resugaring classes, 22/22 hand-written buggy `.lfq` programs, and 240/240 Redex-mutated modules spanning 25 error codes, diagnostics achieve 100% user-span anchoring with zero leaked gensyms or raw wire IDs.

6.4 RQ4: Mechanized Metatheory in Lean 4.35.0

Every formal claim in this paper is mechanized in Lean 4.35.0 across 84 standalone proof modules comprising 2,335 theorems with zero `sorry`, zero `axiom`, and zero wildcard match arms:

- **Type System, Elaboration, and ZX e-Graph Lowering:** Duoidal $5^4 = 625$ capability interchange, QSub preservation and four-way independence, ANF-to-XAG escalation and $O(1)$-query k-inductive loop-invariant soundness, QPhysGrade Bellman–Knuth optimality, non-Clifford ZX parity-phase spider cancellation and $16T \to 8T$ twin-CCZ fusion, QBlock Chebyshev ancilla compression, and three-span diagnostic resugaring hygiene.
- **Functional Data Structures and Domain Algorithms:** Exact $\deg_{\mathbb{F}_2}(f) \leq 2$ MBU characterization, h+1 tree pebbling, contracts and complexity bounds for all nine core and four bootstrapped quantum data structures, Hermitian (H_2, H_3, H_4), lifted bivariate ($\mathbb{F}_4^2$), and Suzuki ($S(8)$) AG Early-MBU DQI exactness, code-distance step-down, and convex-M^* distributed QRNS + GHZ cat-disentangling.

Appendix D indexes the exact Lean 4.35.0 declaration names machine-checked against this manuscript.

6.5 RQ5: Do Real-World Circuits Hit λ_{LFQ}'s Side Conditions?

A fair question for any substructural quantum type system [17, 33, 115] is whether its side conditions target hazards that actually arise in practice, rather than exotic corner cases invented to

Table 7. Empirical relevance of λ_{LFQ}'s typing side conditions across nine third-party quantum circuit libraries: 14 real-world failure patterns ruled out statically by projections of $\Gamma_{\text{in}} \vdash t : Q_e(A) \dashv \Gamma_{\text{out}}$.

Circuit Family	Observed Hazard in Third-Party Circuits	Enforcing Rule	Hits
Public-Key Crypto (3)	Kaliski $x = 0$ dirty ancilla; $\mathbb{F}_{13}$ offset; coset overflow	T-DomImg, U2	4
Symmetric & Hash (2)	ReVerC $\chi_5^2 \neq I$ assumption; missing PLU pivot cleanup	T-DomImg, T-BorrowZ	4
Code ISD & DQI (2)	Singular QGJE submatrix leak; wt-> t syndrome erase	T-DomImg, T-CounitU4	3
LCU / QSVT Walks (2)	Deg-2 Chebyshev shortcut on $U^2 \neq I$; PREP† sign flip	T-SubNorm, T-PhaseClean	3

showcase the formalism. To answer this question empirically, we checked representative arithmetic, symmetric-cipher, decoding, and block-encoding circuits drawn from nine published open-source quantum libraries [33, 34, 40, 48, 49, 58, 60, 89, 98, 120] against the projections of $\Gamma_{\text{in}} \vdash t : Q_e(A) \dashv \Gamma_{\text{out}}$. As Table 7 summarizes (witnesses in Appendix D), third-party implementations violate every logical side condition of λ_{LFQ} across 14 concrete instances—including unguarded $x = 0$ Kaliski modular-inversion garbage (U1), uncollapsed coset carry-rail overflow (U2), missing Bruhat PLU permutation cleanup (T-DomImg, T-BorrowZ), non-involutive $\chi_5^2 \neq I$ reverse uncomputation (U1), singular QGJE submatrix undercounting (U1), and degree-2 Chebyshev walk shortcuts applied when $U^2 \neq I$ (T-SubNorm). Enforcing these side conditions in LakefillQ not only eliminates silent state corruption (e.g., restoring guarded QuantumISD success probability to 94.53%), but directly unlocks the verified algebraic and MBU optimizations evaluated next in §6.6.

6.6 RQ6: Open-Problem Pareto Advances Across Three Execution Regimes

Table 8 highlights LakefillQ's end-to-end algorithmic results across **Single Machine**, **Parallel Machine** ($M^* \leq 256$), and **Serial Multishot** regimes (complete per-mode tables appear in Appendix C):

- **Public-Key Cryptanalysis (ECC-256, RSA-2048/4096):** Composing `PackedBezout` `ShiftZipper` (−512 Q), $s = 4$ `BinomialPebbleSchedule`, Select-SWAP, and totalized Hua squaring solves ECC-256 in **1,088 logical qubits** and 14.34×10^6 **Toffolis** (788 Q in ultra-compact mode), simultaneously beating prior low-width (2,124 Q, 198×10^6 Toffolis [48]) and low-Toffoli (2,871 Q [73]) designs. On RSA-2048/4096, `SkewBinaryCarrySaveNumeral` achieves 1,364 Q / 2,644 Q (896 Q / 1,728 Q under Braun RNS-CRT; 1,924 Q single-focus) while preserving surface-code distance $d = 27$ instead of inflating to $d = 29$. In Parallel Machine mode, zero-carry QRNS + Law-U4 GHZ cat-disentangling cuts optical Bell pairs by 50.0%.
- **Symmetric, Post-Quantum, and DQI:** `DLog-5` and `BP32` S-boxes plus bidirectional half-cipher MBU Grover reach 261/325/389 Q (min-width) and 356 Q (min-gates AES-256); $h + 1$ sequential Merkle pebbling compresses SLH-DSA-128s to 4,320 Q (−34.8%); `StreamingSyndromeAccumulator` + QM4RI/DOOM compresses Classic McEliece-348864 to 7,744 Q (76.26× narrower) and BIKE-L1 to 36,969 Q (4,107× narrower); odd-half NTT compresses ML-KEM-768 / ML-DSA-65 to 3,072/5,888 Q; and coherent OPI DQI shrinks by 10.46× to 9,984 Q.
- **Quantum Chemistry (FeMoCo THC QSVT & Hubbard QPE):** QBlock Chebyshev reflection fusion (192 → 13 ancillas) and Law-U4 PREP† (2,046 → 91 `TempANDs`, 22.48×) lower FeMoCo walk cost to 6,890 `TempANDs`/step [69]. On noisy IBM Heron r2 Hubbard phase estimation [55, 71], Serial Multishot `reset_all` achieves 0.0795 Ha energy error and Parallel Machine achieves 0.0053 Ha, compared to 0.7540 Ha under monolithic Single Machine decoherence.

Table 8. Selected open-problem results compiled by LakefillQ vs. published state of the art (extended Single, Parallel, and Serial Multishot tables in Appendix C).

Problem	SOTA Width	LFQ Width	SOTA Cost	LFQ Cost	Key Gain
ECC-256	2,338 Q (2,124) [48]	1,088 Q (788 min)	197.00M Tof	14.34M Tof (13.74×)	2.15× Q, 13.8× Tof
RSA-2048	4,098 Q (2,052) [24, 40]	1,364 Q (896 RNS)	$d = 29$ seq.	Preserves $d = 27$	344,304 qLDPC Q
AES-256	512–1,376 Q [58, 70, 120]	356 Q (389 min)	40+ AND/S-box	32 TempAND/S-box	Bidirectional MBU
SLH-DSA	6,624 Q Merkle	4,320 Q	$2h + 1$ spines	$h + 1$ two-pass	−34.8% width
McEliece	590,564 Q [34]	7,744 Q	Dense r^2 QGJE	Stream + QM4RI	76.26× narrower
OPI DQI	104,448 Q [60]	9,984 Q	Unitary BM	Bézout + pebbles	10.46× narrower
FeMoCo	192 walk anc. [69]	13 walk anc.	2,046 PREP†	91 PREP† (6,890)	22.48× PREP†

7 SCOPE BOUNDARIES, RELATED WORK, AND CONCLUSION

7.1 Negative Scope Boundaries: When Flat Imperative Circuits Prevail

Purely functional quantum data structures do not improve every quantum circuit; LakefillQ's cost model explicitly identifies three regimes where flat imperative circuits prevail:

(1) **High-Readout-Noise Hardware ($\epsilon_{\text{meas}} > 2\epsilon_{2Q}$) and Slow External Decoders ($\tau_{\text{dec}} > \tau_{\text{cycle}}$).** On devices with noisy dispersive readout or slow camera readout, mid-circuit MBU underperforms measurement-free 3-CZ Margolus RCCX or native RydbergCCZ; QPhysGrade automatically switches away from MBU in this regime (§5.1).

(2) **Small Static Lookup Tables ($M \leq 32$).** Below $M = 32$ entries, Batcher bitonic and crit-bit comparator overhead exceeds a flat Select-SWAP multiplexor.

(3) **QRAM-Free Asymptotic Query Lower Bounds.** Without speculative hardware QRAM, functional data structures reduce circuit width and constant-factor gate counts without violating query lower bounds. In particular, for ML-KEM and ML-DSA lattice sieving, MonoidalFingerTreeFilter cuts inner BKZ similarity checks to 255 Toffolis, yet QRAM-free Select-SWAP candidate scanning imposes $\Theta(N_{\text{vec}}^{1.5}) = 2^{0.31125\beta} > 2^{0.2920\beta}$ classical sieving [10, 23, 59]—honestly yielding a 0× asymptotic quantum sieving speedup without QRAM, just as $O(\kappa \log(1/\varepsilon))$ QSVT [42] and $\Theta(\sqrt{VM})$ minimum-spanning-tree bounds remain binding.

7.2 Related Work

Table 9 compares LakefillQ against representative quantum programming languages. Silq [17] pioneered automatic unitary uncomputation via syntactic qfree/mfree annotations, and Qrisp [102] automates unitary uncomputation via dependency graphs, but neither tracks bilinear X-basis MBU phase kickbacks (U4), domain-restricted algebraic inverses (U1), or hardware decoder/crosstalk grades (Φ_1–Φ_3). Twist [115] verifies entanglement purity (pure), and Tower [114] introduces pointer-based quantum radix trees with unitary pointer uncomputation; LakefillQ complements Tower with pointerless implicit trees, skew-binary carry-save numerals, packed Bézout zippers, and zero-Toffoli MBU phase tracking. Qunity [108] unifies control and Kraus semantics algebraically without substructural MBU or hardware grades, while Cobble [33] types continuous block-encodings without Law-U4 PREP† erasure or involutory Chebyshev ancilla fusion. Quipper [45], QWire [91], Proto-Quipper-R/QuRA [26, 27, 100], VOQC [51], Qualtran [49], and Guppy/HUGR [66] provide linear circuit description, affine width analysis, verified peephole optimization, or resource estimation, whereas LakefillQ integrates duoidal wire capabilities, XAG/FRAIG + Z3 verification, non-Clifford ZX e-graph extraction, and multi-QPU codegen in one mechanized calculus.

Table 9. Comparison of LakefillQ with prior quantum programming languages and verified compilers.

Language / System	Auto Uncomp.	M_X (U4)	Canon. (U2)	HW (Φ)	QBlock	Mechanized
Silq [17] / Qrisp [102]	Unitary	No	No	No	No	Paper / None
Twist [115]	Manual	No	No	No	No	Paper + Coq
Tower [114]	Pointer unit.	No	Radix tries	No	No	Paper
Qunity [108]	Algebraic	No	No	No	Kraus only	Paper
Proto-Quipper-R [27]	Affine width	No	No	Width only	No	Paper
VOQC [51]	Gate pass	No	No	Gate count	No	Coq
Guppy [66] / Qualtran [49]	Linear / Bloq	Dyn. / Est.	No	Bloq counts	Unfused	Rust / Python
Cobble [33]	Block-enc.	No	No	Subnorm. α	Unfused $d{\cdot}m$	Paper
LakefillQ (This Work)	**Unitary+MBU**	**XAG+Z3**	**Yes**	**Φ_1–Φ_3+ZX**	**m_{anc}+1+U4**	**2,335 Lean 4**

Within graded and modal type theory [1, 2, 8, 25, 37, 80, 86, 87, 111, 113], λ_{LFQ} combines uniqueness/linearity duality [77] and place-indexed distributed modalities [107] with two distinctly quantum phenomena absent from classical coeffect calculi: non-additive clean-ancilla reuse along chordal SSA graphs and nonlinear modal actions $r \triangleright e$ capturing Chebyshev phase drift and Zalka's $\sqrt{M}$ search penalty.

7.3 Conclusion

Purely functional data structures and fault-tolerant quantum circuits share a deep structural bond: both achieve efficiency by confining state changes to a small focus or logarithmic spine rather than mutating aliased memory in place. By uniting duoidal wire capabilities ($0 < \mathsf{BorrowZ} < \mathsf{TaintedX} < \mathsf{Lin1} < \mathsf{Uniq1}$) and physical hardware grades in a single compositional judgment $\Gamma_{\text{in}} \vdash t : Q_e(A) \dashv \Gamma_{\text{out}}$, LakefillQ enables annotation-free verification of MBU phase cancellation, non-Clifford ZX e-graph lowering, and Pareto-optimal circuits across cryptography, algebraic-geometry DQI, post-quantum security, and quantum chemistry.

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A HOST-LANGUAGE REALIZATIONS OF QUANTUM DATA STRUCTURES

While `#lang lakefillq` serves as our primary functional surface language in the main body, researchers integrating low-width quantum circuits into existing classical toolchains often embed reversible circuit combinators in typed host languages [3, 36, 97, 103]. Following McBride's derivative view of one-hole contexts [79] and Okasaki–Gill Braun indexing [85], our artifact provides verified host embeddings in Rust (`no_std` gate-level emitter benchmarking Cuccaro [28] vs. Gidney [38] adders), Haskell (`-XLinearTypes`), OCaml functors, and Lean 4.35.0. Listings 7 and 8 illustrate the gate-level Rust MBU ripple-carry primitive and the linear Haskell/OCaml/Lean type signatures corresponding to $\Gamma_{\text{in}} \vdash t : Q_e(A) \dashv \Gamma_{\text{out}}$.

Listing 7. Gate-level Rust `no_std` emitter for Gidney's degree-2 MBU ripple-carry adder ($W - 1$ forward Toffolis, 0 reverse Toffolis).

```
pub fn gidney_ripple_add_in_place_mbu<const W: usize>(
    a: &mut QReg<W>, b: &mut QReg<W>, carry: &mut QReg<W>,
) {
    CCX(a[0], b[0], &mut carry[0]);
    for i in 1..(W - 1) {
        CNOT(carry[i - 1], &mut a[i]);
        CNOT(carry[i - 1], &mut b[i]);
        CCX(a[i], b[i], &mut carry[i]);
        CNOT(carry[i - 1], &mut carry[i]);
    }
    CNOT(carry[W - 2], &mut b[W - 1]);
    CNOT(a[W - 1], &mut b[W - 1]);
    for i in (1..(W - 1)).rev() {
        CNOT(carry[i - 1], &mut carry[i]);
        if MEAS_X_RESET(&mut carry[i]) { CZ(a[i], b[i]); }
        CNOT(carry[i - 1], &mut a[i]);
    }
    if MEAS_X_RESET(&mut carry[0]) { CZ(a[0], b[0]); }
}
```

Listing 8. Linear Haskell (`-XLinearTypes`) GADT signature encoding the five-point wire capability scale and scoped focus-zipper invariant.

```
{-# LANGUAGE DataKinds, GADTs, LinearTypes #-}
data Cap = Cap0 | BorrowZ | TaintedX | Lin1 | Uniq1
data QWire (k :: Cap) where MkWire :: Int -> QWire k
```

```
withFocusZipper :: (QVec w 'Lin1 %1 -> QVec c 'BorrowZ %1 -> (QVec w 'Lin1, QVec c 'BorrowZ))
                -> QVec w 'Lin1 %1 -> QVec c 'Lin1 %1 -> (QVec w 'Lin1, QVec c 'Lin1)
```

B PROOFS OF FORMAL PROPOSITIONS, LEMMAS, AND THEOREMS

We summarize the mathematical proofs of all formal statements in the main paper; every proof is machine-checked without axioms in Lean 4.35.0 (`proofs/*.lean`, indexed in Appendix D).

Proof of Proposition 4.1 (Degree-2 Boundary of Clifford MBU). Applying H and Z-basis measurement $m \in \{0,1\}$ to $|f(x)\rangle$ maps $\sum_x \alpha_x |x\rangle|f(x)\rangle \mapsto \frac{1}{\sqrt{2}} \sum_x \alpha_x (-1)^{m \cdot f(x)} |x\rangle|m\rangle$. When $m = 1$, canceling $(-1)^{f(x)}$ requires a diagonal unitary $D = \mathrm{diag}((-1)^{g(x)})$. By the Clifford hierarchy characterization of diagonal gates [7], a diagonal operator over n qubits is generated by $\{\mathrm{Z}, \mathrm{S}, \mathrm{CZ}\}$ if and only if its phase exponent has polynomial degree ≤ 2 over $\mathbb{Z}_4$, which for Boolean $\{\pm 1\}$ phases $(-1)^{g(x)}$ holds iff $\deg_{\mathbb{F}_2}(g) \leq 2$. When $\deg_{\mathbb{F}_2}(f) \geq 3$ (minimally $f(x) = x_1 x_2 x_3$ on $n = 3$), the minimum Hamming distance from any cubic Boolean function to the second-order Reed–Muller code $\mathcal{RM}(2,3)$ is $2^{3-3} = 1$ out of 8 truth-table entries, bounding the inner-product fidelity $|\frac{1}{8} \sum_{x \in \{0,1\}^3} (-1)^{f(x) \oplus g(x)}|^2 \leq (6/8)^2 = 9/16 = 0.5625$. □

Proof of Proposition 4.2 (Sequential Two-Pass Tree Pebbling). Let $R(h)$ denote the minimum W-qubit digest registers to compute the root of a complete height-h Merkle tree. Simultaneous evaluation computes left child ($R(h-1)$), holds its W-qubit output while computing right child, and hashes into a parent register, requiring $2h + 1$ registers across simultaneous active spines. Factoring the parent sponge/hash absorption into S_{left} followed by S_{right} computes the left child in $R(h-1)$ registers, XOR/absorbs it into the persistent parent state, and uncomputes the left child completely back to $|0\rangle$ before evaluating the right child in the *same* $R(h-1)$ pool, yielding $R(h) = \max(R(h-1) + 1, R(h-1)) = h + 1$ [68, 104]. □

Proofs of Propositions 4.3–4.11 (Quantum Data Structures). (4.3) Rule T-FocusZipper restricts mutable Lin1 capabilities to the w-bit focus, so zero Toffolis touch the $(C-1)w$ BorrowZ wires; constant shifts permute compile-time wire indices (0 gates), and planar layout places cold limbs in a $\sqrt{Q_{\text{cold}}} \times \sqrt{Q_{\text{cold}}}$ memory block bordered by $2\lceil\sqrt{Q_{\text{cold}}}\rceil$ routing tiles. (4.4) Writing $X = \sum_{k=0}^{C-1} (L_k + d_k 2^w) 2^{kw}$ with $L_k \in [0, 2^w - 1]$ and $d_k \in \{0,1,2\}$ preserves integer value modulo 2^{Cw} under window carry folding $d'_k = d_k + c_{\text{out}} \leq 2$ (maintained via $O(1)$-depth inter-window local carry fold), and a single terminal ripple sweep sets all $d_k \mapsto 0$ bijectively. (4.5) By the Braun invariant $0 \leq |L| - |R| \leq 1$, even/odd index parity fixes subtree routing statically without pointer qubits; associative binary folds execute in $\lceil \log_2 k \rceil$ rounds using $(k-1)w$ forward TempANDs. (4.6) Throughout Kaliski/Bernstein–Yang binary GCD [15, 61], $|u| \cdot |g| + |v| \cdot |f| = p$, ensuring $\mathrm{len}(u) + \mathrm{len}(f) \leq N + 1$; packing u MSB-down and f LSB-up in one N-bit register (plus focus guard bits) avoids overlap, half-domain cofactor folding recovers $a^{-1} \bmod p$ bijectively on $\mathbb{F}_p^*$ (checked across all 65,520 slopes of $\mathbb{F}_{65521}^*$), and indicator corrections on $x \in \{0,1\}$ make Hua squaring total on $\mathbb{F}_p$. (4.7) Standard induction on (s,d) [12, 67] gives $L(s,d) = L(s,d-1) + L(s-1,d) = \binom{s+d}{s}$; at $s = 4, d = 4$, $\binom{8}{4} = 70 \geq 32$ Jump-16 windows in 64 qubits and 109 window evaluations. (4.8–4.11) Monoidal prefix associativity validates 255-Toffoli early rejection in Two-Level VTAA; row-pivot tracking in $r\lceil \log_2 r \rceil$ bits makes Bruhat LU total across all 20,160 matrices in $\mathrm{GL}_4(\mathbb{F}_2)$ while column streaming avoids r^2 allocation [92]; Morrison crit-bit splits depend only on highest differing key bits, yielding permutation-invariant quantum states [117] with 14,080-Toffoli amortized parallel-walker bitonic batches (14,417,920 total vs. 23,855,104 Bentley–Saxe online Toffolis); and Walker–Vose bucket weights equal p_i after 1-comparator swap on $2w + 2\mu + 1$ qubits. □

Proofs of Theorems 3.1–3.4, Lemmas 5.1–5.2, Proposition 5.5, and Corollary 5.4. (3.1) Structural induction on all 20 surface constructors eliminates every macro into $\lambda_{\mathrm{LFQ}}^{\mathrm{kern}}$ (5/5 positive reference `.lfq` programs plus Hermitian, lifted bivariate, and Suzuki AG DQI compile cleanly, 7/7 negative violation programs are rejected, 3/3 hardware-violating programs are restored by `repairPhysGrade`, and 1,000 randomized repair instances converge); chordal graph coloring [46] assigns disjoint physical qubits to overlapping live intervals (compressing Boyar–Peralta SSA width from 178 virtual wires with 135 wire reuses). (3.2) Under U1–U4, all scratch wires terminate in $|0\rangle$ on $\mathcal{D}$ and net M_X phase polynomials vanish identically in $\mathbb{F}_2[m,x]/I(\mathcal{D})$ (verified across 4/4 multi-round cipher bug injections in 3,057.9 ms on 10-round AES-128, 4,249.6 ms on 14-round AES-256, and 9,315.4 ms on 18-round Keccak-$f[200]$, matching NIST MixColumns `8e4da1bc`, as well as $O(1)$-query k-inductive loop-invariant obligations across $R \le 1,000$ rounds); thus the reduced output density matrix is a rank-1 pure state $\mathrm{Tr}(\rho_{\mathrm{out}}^2) = 1$. (3.3, 5.1, 5.2) Verified across 12,096 semiring triples, sequential grade addition and parallel max/union preserve $\preceq$, enabling optimal DAG Bellman–Knuth extraction; under 90% heralded erasure, minimum uncorrectable weight satisfies $2t_P + t_E \ge d$. (5.3, 5.4, 5.5) Solving $V_1 c(p/p_{\mathrm{th}})^{(d_1+1)/2} \le \delta$ against $V_0 c(p/p_{\mathrm{th}})^{(d_0+1)/2} \le \delta$ yields $d_0 - d_1 \ge 2k$; for group-algebra BB-qLDPC matrices over $\mathbb{F}_2[x,y]/(x^\ell - 1, y^m - 1)$, $T_{a,b}$ multiplies both blocks by $x^a y^b$, commuting with polynomial check matrices across all 144 shifts of $[[288, 12, 18]]$. (5.6, 3.4) In Early-MBU DQI on $H_2(\mathbb{F}_4)$ (and analogously on lifted bivariate $\mathbb{F}_4^2$, $H_3(\mathbb{F}_9)$, Suzuki $S(8)/\mathbb{F}_8$, and $H_4(\mathbb{F}_{16})$), for any single-symbol error $e_j \in \mathbb{F}_4^*$ at rational point $P_j = (x_j, y_j)$, the syndrome satisfies $(s_0, s_1, s_2) = (e_j, x_j e_j, y_j e_j)$, so $(s_1 \oplus x_j s_0 = 0) \wedge (s_2 \oplus y_j s_0 = 0)$ fires exclusively at index j and `cond-cz` cancels $(-1)^{\mathrm{Tr}(m_j e_j)}$ across all 25 coset leaders (matching 65.22% BCH$(15,7,5)$ and 82.98% OPI width reductions, with 0.7338 Heron r2 routed hardware fidelity). In `QBlock`, when $U^2 = I$, alternating reflections on $|0\rangle^{\otimes m_{\mathrm{anc}}}$ act within invariant 2×2 Jordan subspaces [42] using a single signal ancilla ($m_{\mathrm{anc}} + 1$, 13 anc. on FeMoCo $d = 16$, 91 `TempANDs` = 22.48× PREP† reduction, OpenQASM 3.0 T_2 max error 9.98e-16). □

C EXTENDED HARDWARE, DECODER PACING, AND THREE-MODE TABLES

C.1 External Sub-Microsecond Decoder Pacing Feasibility

In the main body (§3.3), Grade Φ_1 abstracts the QEC control plane solely through a target-profile latency parameter τ_{dec}, keeping LakefillQ independent of custom classical hardware. To confirm empirically that $\tau_{\mathrm{dec}} \le \tau_{\mathrm{cycle}} = 1{,}000$ ns is achievable under bursty MBU traffic without altering quantum processor hardware, we evaluated an external 500 MHz multiplierless Braun-tree + Shiloach–Vishkin Union-Find syndrome decoder (7,700 LUT6, 0 DSP48, $F_{\max} = 614.25$ MHz, 340 ns cryo-SerDes / 420 ns Heron r2 framing, 660 ns cycle slack, 79.55% sparse fast-path bypass). Across 80,000 Stim QEC cycles, unpaced software MWPM ($\tau_{\mathrm{dec}} = 4{,}500$ ns) accumulates a 146,205.0 ns FIFO backlog and drops $d = 5$ Heron r2 logical survival from 0.9973 down to 0.5700, whereas any external decoder satisfying $\tau_{\mathrm{dec}} \le \tau_{\mathrm{cycle}}$ maintains 0.0 ns backlog.

C.2 Extended Cryptographic, Post-Quantum, and Three-Mode Benchmarks

Across public-key factoring and discrete logarithms [24, 32, 40, 41, 44, 64, 95], LakefillQ spans:

- **RSA-2048 and RSA-4096:** Single-focus coset zipper (1,924 Q), Braun-tree RNS-CRT (896 Q RSA-2048; 1,728 Q RSA-4096), and split-register skew-binary carry-save (1,364 Q RSA-2048; 2,644 Q RSA-4096). Decoupling limb carries preserves surface-code distance $d = 27$ on RSA-2048 (891,348 yoked SC physical qubits in 3.19 superconducting days or 24.9 transversal neutral-atom days, and 344,304 BB-qLDPC physical qubits, vs. 1,755,670 yoked SC qubits under $d = 29$ ripple-carry inflation), while RSA-4096 maps to 1,843,882 yoked SC or 583,248 BB-qLDPC physical qubits.

- **ECC-256:** Packed Bézout + Jump-16 pebbles (1,088 Q, 288,216 BB-qLDPC physical qubits, 11,657,520 signed odd-window `TempANDs`; 17,280 single-module Q102 ions vs. 10,490 ions under $K = 4$ serial multishot folding).
- **Symmetric, Hash, Lattice, Code-Based, and Bootstrapped Container Workloads:** Pareto in-place Boyar–Peralta AES-128/192/256 (290 / 354 / 418 Q, 392 Q pure in-place, 17,000 AES-256 `TempANDs`), `DLog-5` min-width (261 Q / 325 Q / 389 Q, 26,264 AES-128 Grover Toffolis), bidirectional half-cipher MBU Grover (6,527 AES-128 `TempANDs`, 7,392 Block-2 cone `TempANDs`), SLH-DSA-128s/256s (4,320 / 4,064 Q with 41.56% constant-lane sponge folding), Classic McEliece-348864 (7,744 Q, 814,336 forward Toffolis, 5.24× QM4RI speedup), BIKE-L1 (36,969 Q, 109.24× DOOM rotational amortization), ML-KEM-768 (3,072 Q, 322,560 Toffolis), ML-DSA-65 (5,888 Q, 1,224,704 Toffolis), OPI DQI (104,448 → 9,984 Q), and the four bootstrapped containers (`NestedDyadicWalkerAliasBox` for THC LCU state preparation, `BrodalOkasakiBootstrappedSkewHeap` for bounded-depth ISD candidate queues, `CombinadicsSkewRALSimplex` for constant-weight code-word indexing, and `OkasakiBinomialCollimationQueue` for Kuperberg dihedral sieve buckets).

Table 10 summarizes LakefillQ's three-mode execution results across Single Machine, Parallel Machine ($M_{\max} = 256$ under Rule T-ParGuard, verified on `distributed_qrns_shor_u4.lfq` with zero reverse ebits and noisy optical DEJMPS distillation at $d = 3, p = 0.05$ consuming 2.11M raw pairs across 131 steps), and Serial Multishot regimes (including noisy Heron r2 Hubbard QPE: 0.7540 Ha Single Machine error vs. 0.0053 Ha 4-QPU Parallel Machine and 0.0827 Ha sampled / 0.0795 Ha exact Serial Multishot at 6.10× per-shot depth reduction).

Table 10. Three-mode compilation summary across Single Machine, Parallel Machine ($M^* \in \{1..256\}$, $M_{\max} = 256$), and Serial Multishot execution.

Workload	Single Q	Per-Run `TempANDs`	Optimal $M^* \times Q_{\text{loc}}$	Optical Ebits	Multishot / Notes
AES-128 Grover	261	6,527	1×261 (Zalka)	0	Bidirectional MBU
ECC-256 ECDLP	1,088	614,040	4×312	9,744 logical	10,490 Q102 ions ($K = 4$)
RSA-2048 Shor	1,364	36,875,000	6×268 (QRNS)	934,200 raw	0-carry QRNS + U4 GHZ
RSA-4096 Shor	2,644	142,812,500	8×356 (24×228)	0 reverse ebits	Knee at $Q_{\text{cap}} = 230$
FeMoCo THC QSVT	2,142	85,156,250	$64 \times 2,142$ (shots)	0 (shot-par.)	6,890 `TempAND`/walk

D EMPIRICAL SIDE-CONDITION WITNESSES, MUTATION MATRIX, AND LEAN 4.35.0 INDEX

Side-Condition Validation Witnesses and Seven Projected Obligations. Seven general static obligations projected from $\Gamma_{\text{in}} \vdash t : Q_e(A) \dashv \Gamma_{\text{out}}$ (Tables 11 and 12) rule out all 14 concrete correctness hazards observed across the nine third-party quantum circuit libraries in §6.5, including 20/32 corrupted basis rows from assuming $\chi_5^2 = I$ in ReVerC, missing PLU permutation uncomputation (4.23 ms XAG check), 7.36× QuantumISD skip-rightmost undercounting (94.53% guarded Aer success probability), and 1.1796 spectral error from applying the degree-2 Chebyshev walk shortcut when $U^2 \neq I$ in Qualtran THC. Full diagnostic and Redex fuzzer artifacts are recorded in `lakefillq_holistic_language_cohesion_report.json` and `lakefillq_redex_surface_diagnostic_fuzzer_report.json`.

Lean 4.35.0 Mechanization Index (2,335 Theorems Across 84 Modules). Every formal claim in this paper maps to zero-`sorry`, zero-`axiom` Lean 4.35.0 theorems verified by the Lean kernel:

Table 11. Per-family empirical witnesses showing how third-party circuit implementations violate λ_{LFQ}'s side conditions.

Library Family	Violated Assumption	Empirical Failure Witness	Rule
Public-Key (3)	$x = 0$ inv.; $\mathbb{F}_{13}$ offset; coset	Dirty ancilla / wrap-around	T-DomImg, U2
Symmetric (2)	$\chi_5^2 \neq I$ (20/32 rows); PLU perm	Corrupted state (4.23 ms catch)	T-DomImg, T-BorrowZ
ISD / DQI (2)	Singular QGJE (7.36×); wt-> t	Amplitude leak (94.53% fixed)	T-DomImg, T-CounitU4
LCU / QSVT (2)	$d = 2$ walk on $U^2 \neq I$; PREP†	1.1796 op. error; LCU cancel	T-SubNorm, T-PhaseClean

Table 12. The seven coordinate projections of $\Gamma_{\mathrm{in}} \vdash t : Q_e(A) \dashv \Gamma_{\mathrm{out}}$ in LakefillQ and the concrete circuit hazards each projection statically rules out.

Judgment Projection	Static Obligation Enforced	Circuit Hazards Ruled Out
T-DomImg (U1)	Domain-restricted $g(x, f(x)) = x$ on $\mathcal{D}$	Kaliski $x = 0$, ReVerC χ_5, singular QGJE, DQI
T-BorrowZ	Read-only BorrowZ context non-interference	In-place PLU alias, cold-limb overwrite
T-PhaseClean (U2)	Canonical discharge & $\Phi_X \equiv 0$ before H/QFT	Uncollapsed carry rail, PREP† sign flip
T-CounitU4 (U4)	Bilinear $\mathbb{F}_2$ M_X phase cancellation at exit	Omitted cond-cz, remote GHZ phase leak
T-SubNorm (QBlock)	$U^2 = I$ involutory guard & α, ε bounds	Qualtran $d = 2$ Chebyshev walk corruption
T-QubitCap (Φ_1–Φ_3)	Hardware capacity, τ_{dec} pacing, exclusion	Lindley backlog, blockade crosstalk, hook
T-ParGuard ($r \triangleright e$)	Convex M^* selection with Zalka $\sqrt{M}$ penalty	Suboptimal Grover/ISD multi-QPU splitting

Table 13. Compiler soundness mutation testing: all 12/12 (100.0%) injected compiler bugs are killed by simulator counterexamples.

Mutant Group	Disabled Compiler Check	Triggering Counterexample	Killed
M1–M4 (QSub)	Disable U1–U4 logical checks	$\mathrm{Tr}(\rho^2) \leq 0.50$ / norm leak	4/4
M5–M6 (SSA/XAG)	Ignore chordal overlap; skip XAG cut-sweep	Recycled-wire clobber; 10-rd taint	2/2
M7–M9 (QPhysGrade)	Ignore Φ_1 τ_{dec}, Φ_2 blockade, Φ_3 hook	Backlog divergence; $d_{\mathrm{eff}} : 5 \to 3$	3/3
M10–M12 (QBlock/M^*)	Skip $U^2 = I$; drop Zalka $\sqrt{M}$; drop GHZ U4	1.18 spec. err.; Bell-cut blowup	3/3

- **Core Calculus & Duoidal Capabilities (§3, Thms. 3.1–3.3, 3.4):** `lakefill_qsub_qphys_architecture_master_closure_theorem`, `asymmetric_within_apply_mbu_halves_t_count_zero_phase_taint`, `thm22b_f28_chordal_register_packing_clean_ancilla_safety`, `elaborateToKern_zero_residual_forms`, `openqasm3_mbu_feedforward_sound`, `repairPhysGrade_restores_wellTyped`, `rwire_ordered_duoidal_interchange`, `rwire_borrow_then_move_safe_vs_par_conflict`, `deltaZ_ccx_and_tempAnd_contraction_sound`, `sequential_callQ_chordal_scratch_saves_wires`, `auto_cluster_profile_eliminates_cost_drift`, `qsub_logical_preservation_and_unit_purity_soundness`, `wire_level_symbolic_type_inference_module_closure`, `bilinear_xag_mbu_collapses_arbitrary_degree_to_zero`, `multiround_mbu_session_total_quantum_phase_unit`, `cut_sweeping_local_unsat_implies_global_multiround_soundness`, `anf_to_xag_escalation_preserves_zero_verdict`, `inductive_loop_soundness_implies_unrolled_qsub`, `inductive_loop_zero_taint_escape`, `inductive_loop_constant_query_bound`, `qphys_grade_seq_associative_and_identity`, `qphys_phi1_phi2_phi3_strict_independence_witnesses`, `lcu_projection_sound_under_u1_u4`, `qsvt_chebyshev_ancilla_compression`, `berryGidney_femoco_L2048_91_lt_2046`, `linTong_serial_multishot_ancilla_and_phi1_advantage`.

- **Uncomputation & Quantum Data Structures (§4, Props. 4.1, 4.2, 4.3–4.11):** degree_guarded_hmr_cz_soundness_and_cubic_rejection, catch1_gidney_mbu_cannot_zero_out_macro_pebble_windows, reversible_tree_pebbling_simultaneous_2h_plus_1_vs_sequential_h_plus_1, api1_huet_focus_zipper_contract_and_cost, api2_skew_binary_carry_save_contract_and_cost, skew_binary_canonical_discharge_preserves_exact_integer_value, api3_braun_tree_fold_contract_and_cost, braun_tree_size_and_lsb_routing_invariant, api4_packed_bezout_shift_zipper_contract_and_cost, hua_identity_with_singular_zero_one_correction_exact, bezout_half_domain_folding_strict_cofactor_bijection, api5_binomial_pebble_schedule_contract_and_cost, api6_monoidal_finger_tree_simhash_contract_and_cost, api7_bruhat_pivoted_lu_and_streaming_syndrome_contract_and_cost, bruhat_lu_f2_zero_pivot_requires_permutation_and_recovers_all_gl4, api8_canonical_critbit_bitonic_dictionary_contract_and_cost, parallel_walker_bitonic_74_47x_amortization_and_width_tradeoff, api9_walker_vose_alias_table_contract_and_cost.
- **ZX e-Graph, Hardware Step-Down, & AG DQI (§5–§6, Lemmas 5.1–5.2, Prop. 5.5, Thms. 5.3–5.6, Cor. 5.4):** zx_spider_fusion_mod8_sound, zx_t_tdg_gadget_cancel, zx_t_t_fuses_to_clifford_s, zx_cx01_parity_transport_sound, shared_twin_ccz_zx_8t_eq_ideal, hermitian_h2_all_points_on_curve, hermitian_h2_exceeds_reed_solomon_bound, hermitian_h2_weight1_syndrome_injective, hermitian_locator_recovers_exact_coset_leader, hermitian_early_mbu_law_u4_phase_cancel, lifted_bivariate_early_mbu_law_u4_phase_cancel, suzuki_s8_genus_and_quadratic_blocklength_law, higher_genus_and_lifted_ag_dominate_rs_mds_ceiling, lindley_zero_backlog_when_paced, lindley_linear_backlog_when_unpaced, external_decoder_pacing_concrete_separation, rccx_paired_within_and_phase_cancellation, knuth_dijkstra_optimal_acyclic_egraph_extraction, four_target_pareto_separation_and_qroam_dominance, bernoulli_scaled_survival_verified, mixed_pauli_erasure_min_fault_weight, erasure_doubles_correctable_faults_all_odd_distances, volume_reduction_steps_down_code_distance_by_2, rsa2048_and_ecc256_code_distance_step_down_verified, f41_s21_cat_qubit_z_bias_preserving_and_76x_compression, dejmps_distilled_epr_photonic_deque_zipper_threshold_soundness, qldpc_bivariate_bicycle_automorphism_zipper_vs_bridge_catch, berry_gidney_sublinear_sqrt_n_temp_and_dominance, kawamura_law_u2_quotient_erasure_and_canonical_representation, early_mbu_phase_cancellation_algebraic_identity, spanning_tree_peel_commutation_Hc_eq_s, lakefillq_grade_phi1_lindley_zero_backlog, unpaced_mwpm_lindley_backlog_divergence.
- **Diagnostics, Fuzzing, Side-Condition Witnesses, & Distributed M^* (§6):** complete_12_of_12_compiler_mutant_kill_rate_soundness, multipass_pipeline_preserves_caller_span, buildProvenanceChain_preserves_user_anchoring, chordal_resugar_recycled_step_blames_second_owner, resugar_meas_eliminates_all_compiler_gensyms, minimal_monomial_witness_evaluates_true, undeclared_wire_rejected_outside_watermark, callQ_aliased_mutated_port_rejected, lisp1_dual_position_operator_preserves_gate, surface_adjoint_involutive, braun_seq_invert_exact_identity, enriched_diag_total_3span_hygiene, ecc_f13_offset_trajectory_sound_vs_reverse_uncompute_hazard, kaliski_zero_input_unguarded_dirty_vs_guarded_clean, coset_carry_runway_non_wraparound_and_tightness, reverc_chi5_not_involution, plu_missing_perm_unsound, sphincs_chi_constant_lane_

eliminates_two_ands_per_row, qgje_non_singular_guard_soundness, mceliece_348864_qm4ri_tempand_reduction, bike_l1_doom_amortized_109x_speedup, qwalk_d2_equals_chebT2_iff_involution, wx_to_refl_phase_d2_sign_flip_cancels_lcu_sum, femoco_thc_law_u4_walk_cost_strictly_dominates_baselines, t_par_guard_zalka_fallback_soundness, t_par_guard_bell_cut_overhead_triggers_single_machine_fallback, law_u4_ghz_cat_disentangle_halves_lookup_ebits, qrns_distributed_zero_carry_ebits_soundness, qpar_m_star_ucurve_knee_pruning_and_monolithic_fit, qpar_m256_horizon_rsa4096_knee24_and_optical_tree_scaling, gf16_inv_4and_obstruction_and_5and_exactness, bidirectional_half_cipher_mbu_grover_dominance, mitm_leaf_gadgets_shor_and_isd_dominance, dlog5_inplace_aes_sbox_256_exact_and_5ancilla_cleanup, ecc256_dual_qec_and_m16_cluster_zalka_contrast, t_par_guard_grover_fallback_soundness, distributed_law_u4_lookup_qrns_and_m_star_soundness.